Technical Appendix: Controller Admission and Helper-Game Preservation

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This appendix supplies the round-to-alternating fixed-point derivation, the source comparisons behind the first-flip counterexample and the published positive-label audit, and the proof and saved-model evidence for cooperative continuation loss. It also reports the eleven published replacement candidates, separate selected-strategy controls, a direct positive witness for Lift, and a controlled extraction intervention followed by a bounded modeled replacement episode. The algebraic result concerns final region values; the Lift extraction and execution checks supply separate, instance-specific evidence.[1]

Reading guide. Section A gives the round-to-alternating derivation; B–C cover source comparisons and archived-label scope. Section D covers saved released arenas and safety-subgame checks; J reports strict-preservation loss calculations, including resource UNKNOWN. For continuation witnesses, E gives Vehicle's saved-model and source evidence, whereas F is source-only and residual. Published-replacement evidence is in G–H; H includes the Lift T diagnosis and modeled episode, and separates Lift P's correspondence failure from its direct graph audit. Section I gives the constructed Choice/Toggle pair. Section K maps the claims to existing saved-evidence archives. Section L gives the full retained-constraint proof and counterexample, cooperation characterization, implementation details, mechanism controls, and detailed decision table. Section L.6 gives the complete literal-preservation proofs, corrected-guard proof, and propagation, initialization, and Guard examples. The main paper states and proves the principal preservation results; this supplement provides expanded derivations, variants, row-level evidence, and reproduction details for closer verification. Section M reports the complete eleven-entry Elevator residual census; the fixed preference was inactive throughout.

A. Round-to-alternating correspondence

Source fixed point and one-edge predecessors. We restate the vector operator of [3, Eq. (15)], using local bound variables and explicit grouping. On the bipartite arena $H = (Q_0, Q_1, \delta_0, \delta_1)$, put $Q = Q_0 \cup Q_1$. For $T \subseteq Q$, its one-edge predecessors are

$$\begin{aligned} \mathrm{Pre}_\exists(T) &= \bigcup_{k=0}^{1} \{q \in Q_k : \delta_k(q) \cap T \neq \varnothing\}, \\ \mathrm{Pre}_1(T) &= \{q \in Q_0 : \delta_0(q) \subseteq T\} \cup \{q \in Q_1 : \delta_1(q) \cap T \neq \varnothing\}, \\ \mathrm{CondPre}_{\mathrm{edge}}(U, R) &= \mathrm{Pre}_\exists(U) \cap \mathrm{Pre}_1(U \cup R), \end{aligned}$$

[1] OpenAI Codex and Anthropic Claude Fable 5.1 assisted with the analysis and drafting; see the main paper's Acknowledgments and AI Assistance for roles and scope.

as in [3, Eqs. (1), (4), and (5)]. For assumption sets $a_1, \dots, a_m \subseteq Q$ and guarantee sets $g_1, \dots, g_n \subseteq Q$, with $m, n \geq 1$, define

$$\begin{aligned}
\Omega_{ji}(\mathbf{Z}, Y, X, U) &= (g_j \cap \mathrm{Pre}_1(Z_{j^+})) \cup \mathrm{Pre}_1(Y) \\
&\qquad \cup \big((Q \setminus a_i) \cap \mathrm{CondPre}_{\mathrm{edge}}(U, X \setminus a_i)\big), \\
\mathcal{T}_j(\mathbf{Z}) &= \mu Y.\, \bigcup_{i=1}^{m} \big(\nu X.\, \mu U.\, \Omega_{ji}(\mathbf{Z}, Y, X, U)\big), \\
\mathbf{Z}^* &= \nu \mathbf{Z}.\, \big(\mathcal{T}_j(\mathbf{Z})\big)_{j=1}^{n}, \qquad j^+ = (j \bmod n) + 1.
\end{aligned}$$

Here the outer greatest fixed point is simultaneous; Y is local to component j, and X, U are local to branch (j, i). The source defines the final region as $\bigcup_{j=1}^{n} Z_j^*$ [3, Appendix B.1]. These operators take one alternating edge. The following lemma instead substitutes the completed-round operators C and P for Pre_1 and $\mathrm{Pre}_\exists$, respectively, so its conditional predecessor is $P(U) \cap C(U \cup R)$. The semantic soundness theorem uses its language-inclusion premise and mode/rank strategy, while completeness quantifies over all system strategies [3, Theorem 3 and Appendix B.2]; the following lemma transfers final regions only, not that strategy construction.

Lemma 1 (Round and alternating fixed points)**.** *Let $H = (Q_0, Q_1, \delta_0, \delta_1)$ be a finite total bipartite arena, where $Q_0 = V \neq \varnothing$, environment edges go from Q_0 to Q_1, and system edges go from Q_1 to Q_0.*[2] *Use nonempty assumption and guarantee justice lists, and place every justice label only on Q_0. Replace the one-edge predecessors in the vector formula of [3, Eq. (15)] by the completed-round operators $C(X) = \forall\delta_0\, \exists\delta_1\, X$ and $P(X) = \exists\delta_0\, \exists\delta_1\, X$, with $\mathrm{CondPre}(U, R) = P(U) \cap C(U \cup R)$. Then each final round vector component equals the Q_0 restriction of the corresponding alternating component. Thus their unions have the same completed-state region.*

Proof. We read the third term of the inner body in Eq. (15) as $(Q \setminus a) \cap \mathrm{CondPre}(U, X \setminus a)$, where $Q = Q_0 \cup Q_1$ and $a \subseteq Q_0$ is the selected assumption. For $X \subseteq Q_0$ and $S \subseteq Q_1$, put

$$\begin{aligned}
R(X) &= \{t \in Q_1 : \delta_1(t) \cap X \neq \varnothing\}, \\
A(S) &= \{s \in Q_0 : \delta_0(s) \subseteq S\}, \\
E(S) &= \{s \in Q_0 : \delta_0(s) \cap S \neq \varnothing\}.
\end{aligned}$$

Then $C = AR$, $P = ER$, and the lift $L(X) = (X, R(X))$ is an order embedding that preserves unions. On pairs, the one-edge operators are $\mathrm{Pre}_1(X_0, X_1) = (A(X_1), R(X_0))$ and $\mathrm{Pre}_\exists(X_0, X_1) = (E(X_1), R(X_0))$. Totality gives $L(V) = Q_0 \cup Q_1$.

Fix lifted outer parameters $Z = L(z)$ and $Y = L(y)$, a guarantee g, and an assumption a. Write $N = V \setminus a$, $T = g \cap C(z)$, and $B = T \cup C(y)$. Temporarily assume $y \subseteq B$. For fixed alternating $X = (x_0, x_1)$, the inner least fixed point has equations

$$U_0 = B \cup \big(N \cap E(U_1) \cap A(U_1 \cup x_1)\big), \qquad U_1 = R(y \cup U_0).$$

The second equation follows because justice is absent on Q_1 and $R(U_0) \cap R(U_0 \cup S) = R(U_0)$ for every reserve S. Every fixed point has $U_0 \supseteq B \supseteq y$, so $U_1 = R(U_0)$. Conversely, a fixed point of $u \mapsto B \cup (N \cap P(u) \cap A(R(u) \cup x_1))$ lifts to a fixed point of this paired system. This correspondence preserves and reflects order, hence transports the least fixed point. The resulting outer X map has image in the lifts. Its fixed points therefore correspond, again in both order directions, to those of

$$J(x) = \mu u.\big[B \cup \big(N \cap P(u) \cap C(u \cup x)\big)\big].$$

[2] AI-assisted analysis and drafting: [7, 8].

Consequently the alternating branch is $L(\nu J)$.

The round branch uses instead

$$I(x) = \mu u.\big[B \cup \big(N \cap P(u) \cap C(u \cup (x \cap N))\big)\big].$$

Pointwise $I \leq J$, so $\nu I \subseteq \nu J$. For $x^* = \nu J$, every output of the inner body lies in $B \cup N$, whence $x^* \setminus N \subseteq B$. Each inner fixed point contains B; on every $u \supseteq B$ we have $u \cup x^* = u \cup (x^* \cap N)$. The two inner maps at x^* thus have the same fixed points and the same least fixed point. Hence $I(x^*) = J(x^*) = x^*$, giving the reverse inclusion and $\nu I = \nu J$.

It remains to justify $y \subseteq B$ along the local μY iteration. It holds at $y = \varnothing$. Let x_a be the final branch for assumption a, and put $t = \bigcup_a x_a$. Every x_a contains B, so $y \subseteq B \subseteq t$. At its nested fixed point,

$$x_a = B \cup \big((V \setminus a) \cap P(x_a) \cap C(x_a)\big).$$

Monotonicity therefore gives $t \subseteq B \cup C(t) \subseteq T \cup C(t)$. Thus the required invariant holds for the next Y value. Since R preserves unions, the alternating update is $\bigcup_a L(x_a) = L(t)$. Induction and finite stabilization establish correspondence of the local least fixed points. Finally, the simultaneous outer vector iteration starts at the lifted top vector; the target in component j is $g_j \cap C(z_{(j \bmod n)+1})$ on Q_0 and empty on Q_1, where n is the number of guarantee components. Applying the local argument at every update proves componentwise correspondence at the final outer greatest fixed point. □

Scope. This lemma identifies final fixed-point values; it does not identify all inner approximants or extracted ranks. Appendix B.1 of [3] defines the vector formula's region as the union of its final components, so componentwise correspondence also identifies the completed-state restriction of that union. Before applying the lemma, empty justice lists are padded by V in rounds and by Q_0 in the alternating arena. This gives nonempty index sets on both sides. The padding also preserves the objective because Q_0 is visited infinitely often on every play. If $V = \varnothing$, the region is empty directly; no initialized game theorem is invoked. For the arena H_V, totality follows from $V = \nu X.(\mathrm{Alive}_E \cap \mathrm{CPre}(X))$: δ_0 retains every original-legal input, and δ_1 retains all original-safe responses into V, at least one for each such input. The lemma is algebraic and does not require justice normalization; normalization supplies the premise for the separate semantic soundness theorem.

B. Source comparisons

Comparison with the published preservation proof. The Guard counterexample uses Boolean input x, response y, trivial initial and justice predicates, and $\rho_e = \rho_s = x'$.[3] Section 4.1 of [4, p. 4] writes the converted clause forms as $\rho_n^e \vee \mathrm{next}(\mathit{envHelp})$ for each safety assumption and $\rho_n^s \vee \mathit{envHelp}$ for each safety guarantee. We denote *envHelp* by b and its next value by b', conjoin the respective clauses, and include helper persistence. Theorem 1 of [4, p. 5] takes M' to be the specification obtained by reducing M, and concludes: "Then M is realizable if and only if M' is realizable." The literal reduction has environment safety $\rho_e^R = (\rho_e \vee b') \wedge (b \Rightarrow b')$ and system safety $\rho_s^R = \rho_s \vee b$.

The gap is specifically on the edge into the first helper-true state. Appendix A of [4, pp. 12–13], in the second part of the proof of Theorem 1, reasons that system safety holds after the helper is set to true and writes the relaxed environmental safety using an unprimed helper. Section 4.1 instead guards environmental safety with the *next* helper. On an edge $b = 0, b' = 1$, the input is therefore already released while the system response is not. The later failure of environmental fairness cannot discharge that edge's strict safety obligation. The Sec. 4.1 clause forms as written therefore do

[3] AI-assisted analysis and drafting: [7, 8].

Table 1: Helper variants and applicable strict-preservation results. The first-flip column describes original-variable safety on $b = 0, b' = 1$; $\top$ means released. Complete semantic variants guard all safety constraints. A syntactic route requires complete relation checks after all translations, including monitor updates. These results alone do not establish raw cooperative-core verdict transfer or returned-graph cooperation.

Variant	First-flip ENV / SYS	Applicable result	Evidence in the main paper
Kind [4, Sec. 4.1] literal relations	$\top$ / ρ_s	Main Theorem 1: exact surviving strict region; full proof in Section L.6.	The 59-model study measures literal strict loss. The matching Vehicle arena loses a cooperative continuation while preserving initialization.
Kind [4, App. A] proof notation	ρ_e / ρ_s	Neither safety duty is released on the first flip: the environment guard uses current b. Outside Guard's scope.	Source comparison in Supplement B; not identified with the configured syntactic route.
Complete corrected semantic variant	$\top$ / $\top$	Proposition L.4 in Section L.6: strict-region and initial preservation.	Existing released semantic guard, analyzed as complete semantic relations.
Released syntactic route	Arena-specific	Helper arenas: Supplement L, Proposition L.1, after relation checks. Checked no-helper arenas: original-model identity.	All eight completed helper and five identity cases preserve strict winning. Separately selected Humanoid 044 loses strict initialization; it already lacks cooperative initialization (main Section V).

not satisfy this stated equivalence. Appendix A's different environmental guard and the released semantic correction are outside this counterexample's scope. It does not by itself refute the separate well-separation statement in Theorem 2.

Existing semantic correction. In the released `PitermanSemanticReduction.java`, version `1.0.0.202402072123`, `doReduction` lines 28–29 add `envHelpIsTrue.id()` and `envHelpIsFalse.and(envHelpPrimeIsTrue)` to system safety through two successive `orWith` calls. These are exactly the guards b and $\neg b \wedge b'$ analyzed in the main paper. The same calls appear at lines 56–57 of the saved source snapshot [5]. The method returns without adding a helper when environmental safety is universally true. This source comparison does not identify the configured syntactic arena with the complete semantic variant.

Table 1 compares these forms with the saved syntactic arenas and their applicable strict-preservation results.

Raw recurrence for the Justice control. For the Justice control in Section L.2, D is the typed carrier, $D_0 = \{\neg p\}$, $\rho_e = (e' = p)$, $\rho_s = \neg p$, $A = \mathbf{GF}e$, and $G = \mathbf{GF}\neg p$. The all-zero play satisfies G but violates A, witnessing the missing premise already in the original game. The raw single-justice four-fixed-point recurrence returns D_0: its predecessors satisfy

$$\mathrm{CPre}(T) = \mathrm{EPre}(T) = \neg p \wedge \big(T(0,0) \vee T(0,1)\big),$$

and the outer iteration is $D \to D_0 \to D_0$. With either helper guard, its raw region is $\neg p \vee b$, and its initial answer is true. This is a diagnostic of a missing applicability condition, not a counterexample to the conditional theorem of [3].

PCar464: initial impossibility with auxiliary memory. The source clauses used below are explicit here. Write τ for `throttle`, o for `obstacle`, z for `sideSense`, and q_b, q_f for `spec_b2f`, `spec_f2b`. The declared domains are $o \in \{\mathsf{CLEAR}, \mathsf{BLOCKED}\}$, $z \in \{\mathsf{CLEAR}, \mathsf{P_O}, \mathsf{BLOCKED}\}$, $\tau \in \{\mathsf{BWD}, \mathsf{FWD}, \mathsf{STOP}\}$, and *steer* $\in \{\mathsf{LEFT}, \mathsf{RIGHT}, \mathsf{CENTER}\}$. Both direction monitors start in $S0$ and range over $\{S0, S1, S2\}$. Their exact source updates on these enum values are:

	next q_b			next q_f		
current monitor	BWD	FWD	STOP	BWD	FWD	STOP
$S0$	$S1$	$S0$	$S0$	$S0$	$S1$	$S0$
$S1$	$S1$	$S2$	$S0$	$S2$	$S1$	$S0$
$S2$	$S2$	$S2$	$S2$	$S2$	$S2$	$S2$

Each subtable uses its own current monitor as the row and the current throttle as the column. No claim about illegal bit encodings is needed. For the SYS-owned Boolean response auxiliaries r_B, r_Z, initialization sets both to true, and system safety includes

$$r'_B \leftrightarrow (o = \mathsf{CLEAR}) \vee (r_B \wedge \tau \neq \mathsf{BWD}),$$
$$r'_Z \leftrightarrow (z = \mathsf{CLEAR}) \vee \big(r_Z \wedge \neg(\tau = \mathsf{FWD} \wedge \mathit{steer} = \mathsf{CENTER} \wedge z = \mathsf{BLOCKED})\big).$$

The four environment justices are

$$\mathbf{GF}(q_b = S0 \vee q_b = S1), \quad \mathbf{GF}(q_f = S0 \vee q_f = S1), \quad \mathbf{GF}r_B, \quad \mathbf{GF}r_Z.$$

The physical system clauses require

$$\mathbf{G}(o = \mathsf{BLOCKED} \Rightarrow \tau \neq \mathsf{FWD}),$$
$$\mathbf{G}((o = \mathsf{CLEAR} \wedge z = \mathsf{CLEAR}) \Rightarrow \mathit{steer} = \mathsf{CENTER}),$$
$$\mathbf{GF}(\tau = \mathsf{FWD} \wedge \mathit{steer} = \mathsf{CENTER}).$$

Original environment safety is unrestricted. The response-pattern updates belong to system safety even though their justices belong to the environment. These clauses transcribe the unchanged source `PcarLTL_464_PCar_fixed.spectra`, lines 3–15 and 22–70; Supplement K's C14 entry links it and the saved initializer argument.

PCar also admits a source-level impossibility argument from its declared initializer, independently of the extracted graph and our region computation. Suppose a strict-winning, legal-prefix cooperative strategy exists, with arbitrary private memory. Start with an allowed environment valuation with $\mathit{sideSense} = \mathsf{CLEAR}$, and choose

$$\mathit{obstacle}_{t+1} = \mathsf{CLEAR} \iff \mathit{throttle}_t = \mathsf{BWD},$$
$$\mathit{sideSense}_t = \mathsf{CLEAR}.$$

This policy respects Mealy order and the unrestricted environment safety relation. Cooperation prevents either direction-change monitor from reaching its absorbing state $S2$, because its environment justice would then have no satisfying continuation. For $t \geq 1$, a blocked obstacle prohibits forward

motion. A clear obstacle means the preceding throttle was backward, which puts the backward-to-forward monitor in $S1$; forward motion would then force $S2$. Thus forward motion occurs at most initially, violating the system justice requiring forward motion with centered steering infinitely often.

Nevertheless, every environment justice holds. Both monitors remain outside $S2$, and the side-sensor response holds continuously. For the other response auxiliary, a backward throttle at t produces a clear obstacle at $t+1$ and a true auxiliary at $t+2$. Infinitely many backward choices therefore satisfy its justice. If t is the final backward choice, the auxiliary is true at $t+2$ and remains true thereafter; with no backward choice, its true initializer is preserved. This contradicts strict winning and establishes cooperative initial infeasibility.

Meaning of the published labels. The archived SYN15 coverage CSV records TRUE/TRUE for Elevator386 and PCar464 at physical lines 20 and 33. Its columns are Filename, 4FP, and R+4FP; the README identifies them as realizability results underlying the paper's evaluation. The C-backed coverage driver consumes a Boolean, without a returned-controller certificate or per-run source hash. Section C gives the complete source association and aggregate checks; exact historical translated-model equality remains unestablished.

Released route and later source continuity. We execute release `1.0.0.202402072123`. Its packaged UI bytecode exposes the syntactic Kind reduction as a selectable option and initializes ordinary GR(1) as the default. The separate packaged semantic reduction is not invoked by this selected route; we do not report a Lift T outcome for that alternative. A separate audit at Spectra commit `c1ced71d` [5] found that its Java four-fixed-point core and both helper-reduction classes match the packaged release sources after removing only the leading licence comment and normalizing line endings. At this snapshot, the selectable Kind realizability route invokes the syntactic reduction before default translation and BDD generation, then selects a separated solver for ordinary contracts; ordinary GR(1) remains the initialized default. These facts establish source continuity at this snapshot, without establishing a current-build outcome. The CUDD wrapper and native interface have changed, and the surrounding solver hierarchy also differs. A matching generated-language/parser and dependency build was not qualified for this commit. Frontend provider and translation sources are present; their complete build and execution remain unverified.

C. Published positive sources: attribution and semantic scope

Population and attribution We join the previously fixed cohort of all 59 authoritative independent v2 original-model certificates to the four archived `4fp_cover` tables [6].[4] The cohort was selected by availability of complete certificates, without filtering on region sizes or outcomes. For each CSV filename, we construct its exact archive path in the corresponding SYN suite and require that path to belong to the source unit's recorded alias set. We then verify that the archived bytes agree with the roster, local original source, import closure, and saved model input. This is an exact co-packaged-source association, not a per-run historical input receipt.

The CSVs contain 320 raw records: 48, 74, 12, and 186 in SYN15, SYN17, SYN20, and SYN23', respectively. SYN17 contains one `error` record and SYN20 one `an error occurred` record. The remaining 318 completed Boolean records have R+4FP-positive totals 29/48, 61/73, 10/11, and 93/186, matching Table 1 of [4]. Error records are not negative decisions.

Of our 59 source units, 23 match 25 records and 36 are unmatched. Of the matched units, 18 have positive labels (19 records) and five have negative labels (six records). Source unit 055 has

[4] AI-assisted analysis and drafting: [7, 8].

Table 2: Archived labels versus original initial cooperation on the same 23 saved models. Entries count source units (CSV records). The 11 positive-labelled units with $I_M(K) = \mathsf{F}$ comprise seven independent obstructions and four using the proved correspondence. Two of the eleven (072 and 159) depend on the audited initializer convention. Historical C-model identity is unestablished; this is not an error-rate estimate.

Archived R+4FP	$I_M(K) = \mathsf{T}$	$I_M(K) = \mathsf{F}$
Positive	7 (7)	11 (12)
Negative	0 (0)	5 (6)

two byte-identical aliases with positive records; unit 056 has two aliases with negative records. No matched record is an error. Unmatched units have no classification in these particular coverage tables; they are not published negatives. The seven units in Table 3 each correspond to one positive record. This bounded certified cohort is not a random sample, and the counts support no estimate of error prevalence in the published corpus. Table 2 summarizes the separate original-initial cooperation classification of these 23 matched source units.

Initial predicates and witness counts For a typed region R we use

$$I_M(R) = \forall e : \theta_e(e) \Rightarrow \exists y : \theta_s(e, y) \wedge R(e, y),$$

where y includes every SYS-owned logical auxiliary. The number n_E counts typed ENV valuations satisfying θ_e. The missing count m_R counts those valuations for which no such response exists.

All 59 certificates give $I_M(W) = \mathsf{true}$; 12 give $I_M(W_V) = \mathsf{false}$. Six of those 12 have published positive labels: 032, 059, 060, 072, 159, and 164. They also have $I_M(V) = \mathsf{false}$. The separate PCar464 (058) source argument establishes its initial cooperative obstruction despite $I_M(V) = I_M(W_V) = \mathsf{true}$. The separate K cross-check classifies the same 23 saved models using the proved round-to-alternating correspondence and justice normalization. Of the 18 positive-labelled units, eleven fail $I_M(K)$ and seven pass. Besides the seven obstructions above, PCar variants 054, 055, 061, and 062 pass $I_M(W)$, $I_M(V)$, and $I_M(W_V)$ but fail $I_M(K)$. Their K sizes are 16, 0, 42, and 16, respectively; none of their 2, 2, 6, and 2 allowed initial input choices has an initialized response in K. These four conclusions use the correspondence theorem; the previous seven do not require it. All five negatively labelled units also fail $I_M(K)$. A positive $I_M(K)$ certifies legal-prefix strategy existence in the saved model, not cooperation of a particular graph or the stronger all-history recovery property. The exact-loss calculation for corpus159 remains resource UNKNOWN; an initial obstruction does not fill that missing result. Among the five newly linked positive units (excluding the already linked 032), 059, 060, 072, and 159 belong to the existing ten cases failing $W \subseteq C$; 164 does not.

C.1 RoboticArm232: contradictory next-state assumptions

The literal source `roboticarm_Robot_232.spectra` declares six environment variables $s_0, \ldots, s_5$ with the enumeration $\{\mathsf{RED}, \mathsf{GREEN}, \mathsf{BLUE}, \mathsf{EMPTY}, \mathsf{NOT}\}$ (lines 6–17), and an environment variable *task* with domain $\{\mathsf{SCAN}, \mathsf{SORT}\}$ (lines 13, 20–24). There are no system or auxiliary declarations. The only declared initial assumption fixes $task = \mathsf{SCAN}$ (lines 30–31), so there are $5^6 = 15625$ initial ENV choices and the unique empty SYS response. Setting every $s_i = \mathsf{RED}$ is permitted.

Lines 69–83 define

$$\mathit{strongAssumptions} = \bigwedge_{i=0}^{2} (s_i \neq \mathsf{EMPTY} \wedge s_i \neq \mathsf{NOT}) \ \wedge \ \bigwedge_{i=3}^{5} s_i = \mathsf{NOT}.$$

Table 3: Seven source units with archived R+4FP-positive labels and initial cooperative obstructions. Counts concern the saved original Java models: n_E counts allowed initial ENV choices; m_R counts choices missing a SYS/AUX response in R. All seven have $I_M(W) = \mathsf{T}$. PCar464's obstruction has a separate source proof; RoboticArm232's literal source has no legal successor. The initializer obstructions for 072 and 159 depend on the audited state-invariant translation convention (Section C.4).

Source unit	n_E	m_V	m_{W_V}
032 Elevator386	4	1	1
058 PCar464	6	0	0
059 PCar552	6	6	6
060 PCar553	6	6	6
072 Airport145	10	8	9
159 RoboticArm227	125	124	125
164 RoboticArm232	15625	15625	15625

Line 89 requires $\mathbf{GX}\mathit{strongAssumptions}$. Since $\mathsf{EMPTY} \neq \mathsf{NOT}$, every next s_i must differ from EMPTY. In contrast, line 107 requires $\mathbf{GX}\mathit{colorExists}(\mathsf{EMPTY})$, whose definition at lines 111–117 is $\bigvee_{i=0}^{5} s_i = \mathsf{EMPTY}$. Thus the same next state must contain an EMPTY value and contain none. No current state has an environment-safety-legal successor. In particular, the nonempty declared initial set has no legal infinite continuation, so no strategy can meet legal-prefix cooperation. The additional $\mathbf{G}(\mathit{task}' = \mathsf{SORT})$ assumption does not remove this contradiction. This is a direct propositional argument over the source formulas, independent of a region calculation or controller extraction.

Read-only checks of the saved original-model export provide a consistency check: its parsed and expanded ENV ASTs retain both opposing next-state constraints, its saved ρ_e is false, and its independent certificate has $|V| = |W_V| = 0$. These stored results are not premises needed for the source proof. The exact co-packaged source has a positive record at physical line 46 of the SYN17 coverage CSV (header counted as line 1). This establishes the relation between the reported source label and the literal source contract; it does not reconstruct the historical parser, translation, reduction, C solver state, or returned controller.

C.2 RoboticArm232: a direct raw-core calculation

For the preceding source, $\rho_e = \bot$, $\rho_s = \top$, and both justice lists are empty. Its system-response carrier contains the single empty assignment. The original arena is deadlocked: language inclusion holds vacuously, while nonblocking and cooperation fail.

Add the fresh environment helper b and retain the original initializer with $\neg b$. Both the complete literal guard and its first-edge correction simplify to

$$\rho_e^R = (\bot \vee b') \wedge (\neg b \vee b') = b', \qquad \rho_s^R = \top.$$

The transformed typed carrier D_R has $5^6 \cdot 2 \cdot 2 = 62500$ states, counting the colors, task, and helper. The initializer fixes task and helper, leaving the 5^6 choices above. Every state has legal environment moves and a safe system response; every infinite play has $b = 1$ after its first step. The helper assumption $A = \mathrm{GF}\neg b$ is therefore impossible on this total arena. The unchanged empty system-justice list represents $G = \top$; padding it by one tautology preserves that meaning, as in the released `GR1Game` constructor. Padding is distinct from copying environment justices into the system objective.

Let $a = \{b = 0\}$ and write CPre and EPre for the round predecessors. The singleton raw fixed point of [3, Eqs. (9), (15)] is

$$\nu Z.\, \mu Y.\, \nu X.\, \mu U.\ \mathrm{CPre}(Z) \cup \mathrm{CPre}(Y) \cup \big((D_R \setminus a) \cap \mathrm{EPre}(U) \cap \mathrm{CPre}(U \cup (X \setminus a))\big).$$

At $Z = D_R$, totality gives $\mathrm{CPre}(D_R) = D_R$. Thus the body is constantly D_R, all inner fixed points are D_R, and the outer update fixes D_R. Every declared initial input has its empty system response in that region, so the initialized raw decision is positive. Directly in the alternating formula, a tautological guarantee makes the body the whole arena. With justice labels restricted to completed states, the inner least fixed point reaches those states and then all intermediate states, giving the same result.

Here $L(G)$ is nonempty and $L(A)$ is empty. The cited conditional cooperation guarantee therefore does not apply; this calculation does not refute it. Normalization changes the effective guarantee to a. Since every successor has $b' = 1$, both predecessors of a and of $\varnothing$ are empty. The normalized outer sequence is $D_R, a, \varnothing$. Keeping the redundant tautology as a separate component also yields the empty final region. Original-arena viability instead rejects immediately because $V = \varnothing$; transformed viability is D_R and cannot enforce helper fairness.

This is a hand derivation for the stated semantic pipeline on a public source, consistent with its archived positive label. Exact historical C-model identity and the historical decision's cause remain unknown. No native execution or returned-controller observation is claimed.

C.3 PCar552 and PCar553: a one-edge obstruction

Both sources initialize *parkResult* = NA and *spec_policy* = SEEKING, with *spec_park* = false. Choose the allowed initial inputs *obstacle* = BLOCKED and *sideSense* = BLOCKED. The environment may keep those sensor values and choose *parkResult*′ = PARALLEL. The sensor-freeze assumption is met regardless of the initial throttle, and the only other direct ENV safety assumption is the current-state implication

$$\mathit{spec_policy} = \mathsf{SEEKING} \ \Rightarrow\ \mathit{parkResult} = \mathsf{NA},$$

which holds at the chosen initial state. The response patterns add SYS-owned monitor updates and ENV justices, not a restriction on this next ENV input.

System safety requires *spec_policy*′ = SEEKING when the current obstacle is BLOCKED and the current policy is neither PARKING nor PARKED. Every safe response therefore reaches a completed state with policy SEEKING and parkResult PARALLEL. There the current-state ENV implication is false for every next input, so no legal continuation exists. A response that avoids the forced policy violates strict safety on the ENV-legal edge. Initial SYS choices that violate another current-state guarantee likewise cannot rescue a strict strategy. Thus no strict cooperative strategy exists for all declared initial inputs, including strategies with arbitrary private memory. This one-input argument does not derive the stronger saved missing count of six.

C.4 AirportShuttle145 and RoboticArm227: initializer scope

For AirportShuttle145, the declared ENV initializer requires a closed door and no technician acknowledgement, but leaves station and stopButton unconstrained. Choosing station ONE and stopButton PRESSED falsifies the current-state assumption *atStation* ⇒ *stopButton* = RELEASED, where *atStation* means station is not NONE. Every SYS/AUX initial response therefore has no ENV-safety successor. This explains an initializer obstruction, not all eight missing V choices or the ninth missing W_V choice.

For RoboticArm227, the declared initializer permits task SCAN and any of 5^3 color triples. Its current-state assumption requires SCAN to imply $(s_0, s_1, s_2) = (\mathsf{EMPTY}, \mathsf{RED}, \mathsf{NOT})$. Each of the other 124 triples gives no first ENV-safety successor, independently of all SYS/AUX choices. This explains its saved missing V count 124, but does not establish the additional W_V obstruction for the remaining tuple.

These last two arguments depend on the declared initial predicate remaining separate from current-only ENV safety. A convention that incorporates those invariants into the initial predicate can exclude the witness inputs. This matters for the historical coverage route, whose saved preprocessing reclassifies ENV SAFETY as STATE_INV. The archived StateInvTranslator disassembly has guards for next/prime and for ENV formulas referring to SYS/AUX variables before initial strengthening. The relevant Airport and RoboticArm227 current-only formulas have no SYS/AUX reference, so their witnesses may be removed. Both direct PCar552/553 ENV safety formulas refer to SYS variables; the sensor-freeze formula also contains next. These local observations do not prove equality of the full translated models.

Limits of the historical inference The raw CSVs report Boolean feasibility labels, not cryptographic per-run source identities or returned-controller certificates. The literal-source contradiction for RoboticArm232 remains a source-contract fact even if historical preprocessing changed the model's meaning. We neither infer the cause of its reported positive label nor claim that the historical C-backed model is identical to our saved Java model. The other saved-model obstructions and initializer-sensitive arguments retain their stated scope. No new native, BDD, fixed-point, or controller execution was used for this attribution and written-proof analysis.

Elevator032: complete graph details The checked counts are $|W| = |C| = 12$ and $|V| = |W_V| = |K| = 11$; exactly one of four initial inputs lacks a response in K. The released 14-node graph yields an original view with nine nodes, 32 edges, and all four initial nodes. Initial node 3, $(\texttt{NONE}, \texttt{STOP})$, is a dead end there; its native successors all enter helper-true states. The saved legal prefix $0 \to 9 \to 3$, with valuations $(\texttt{LOWER}, \texttt{STOP})$, $(\texttt{LOWER}, \texttt{FWD})$, and $(\texttt{NONE}, \texttt{STOP})$, also reaches it from another initial node. Restricting initialization alone would therefore not make this existing graph cooperative.

PCar058: complete graph details Its strict and nonblocking regions have $|W| = |C| = |V| = |W_V| = 1404$, but $K = \varnothing$, on a 1944-state carrier. All six initial input choices have a response in W_V. No helper is added: the syntactic pass returns before response-pattern expansion because there is no direct parsed ENV safety constraint. The native graph and original view both contain 96 nodes and 576 edges.

D. Per-input safety-subgame cardinalities

For the eight helper-bearing outputs in the fixed 14-input comparison, D is the full typed original state space, including every logical auxiliary; U is the safety-winning region after the helper is set; and F is the condition for a safe entry into U.[5] Section L proves the decomposition summarized in the main paper; the full relation comparisons follow below. The five other completed outputs have no helper and preserve the original model identically. These rows do not include the separately selected Humanoid 044 or AutonomousVehicle 188.

[5] AI-assisted analysis and drafting: [7, 8].

Complete fourteen-input roster. The subset contains all available independently certified carriers of at most 4,096 states. Selection was fixed before an interrupted second extraction attempt, without selecting by region emptiness or controller outcomes; that later attempt did not change these fourteen inputs. Table 4 lists the fixed inputs used on all four paths. The identifiers and source paths come from the frozen `small_native_recovery/SELECTION.json`; the same fourteen identifiers appear once per path in the 56 saved rows of `PUBLIC_14X4_OUTCOMES.tsv` (Section K). Helper status is taken from the released-route records, with the incomplete SmartHome1 run remaining UNKNOWN. AutonomousVehicle 184 is distinct from the separately selected continuation witness AutonomousVehicle 188.

Table 4: The fourteen public inputs. Source suffixes are relative to `artifact/experiments/specs/`. "Helper" records the released route; UNKNOWN does not mean that no helper was added.

ID	Source suffix	Helper
002	`example/HumanoidLTL_377_Humanoid.spectra`	No
032	`syn15/ElevatorLTL_386_Elevator.spectra`	Yes
034	`syn15/GyroLTLVar3_702_GyroAspect.spectra`	Yes
035	`syn15/GyroLTL_390_GyroAspect.spectra`	Yes
036	`syn15/GyroLTL_486_GyroAspect.spectra`	Yes
037	`syn15/GyroLTL_588_GyroAspect.spectra`	Yes
038	`syn15/GyroLTL_590_GyroAspect.spectra`	Yes
043	`syn15/HumanoidLTL_409_Humanoid.spectra`	No
057	`syn15/PcarLTL_394_PCar.spectra`	Yes
058	`syn15/PcarLTL_464_PCar_fixed.spectra`	No
093	`syn17/AirportShuttle_AirportShuttle_49.spectra`	Yes
149	`syn17/SimpleVehicle_VehicleSpec_6.spectra`	No
184	`syn20/AutonomousVehicle_sprint01_ver1.spectra`	No
223	`syn20/SmartHome1_sprint01_ver1.spectra`	UNKNOWN

Full relation checks All 13 completed released arenas from the fixed 14-input comparison preserve the original strict-winning region. The remaining input keeps its resource UNKNOWN. We establish this by auditing the full saved relations and applying the safety-subgame decomposition proved in Section L, retaining every original typed variable and logical auxiliary.

Eight outputs introduce the helper. For each, the independently calculated safety region is $U = D$ and the safe-entry condition is $F = C$. Seven have $F \subsetneq D$, but every original winning state belongs to F, so $W_F = W$. Direct strict GR(1) solution of each full saved transformed arena agrees with W_F at $b = 0$ and U at $b = 1$. Five no-helper outputs instead satisfy complete original-model identity. All 13 original strict initial decisions remain true, and original regions match their prior independent certificates under complete typed name/value mappings across encodings. Table 5 gives per-input cardinalities.

We compare ENV and SYS safety separately over all typed current/next valuations in $(b, b') = 00, 01, 10, 11$ and on legal joint edges. Elevator matches both complete guards because its original system safety is true. The other seven helper cases match literal ENV safety everywhere and original SYS safety in 00 and 01, but retain nontrivial SYS constraints in 11: neither complete guard matches, even on legal joint edges. SYS differences in environment-impossible 10 are recorded separately. Original initialization and ordered justices, including helper justice, match throughout the typed current domain. Thus the complete-literal theorem alone cannot establish preservation here.

Retained constraints therefore need not cause loss: here they permit indefinite safe responses

Table 5: All eight helper inputs in the fixed 14-input cohort. Full typed original states include every logical auxiliary. Exact function comparison gives $W_R(\cdot, 0) = W_F = W$ and $W_R(\cdot, 1) = U$ in every row. The five no-helper cases also preserve W.

ID	$\lvert D\rvert = \lvert U\rvert$	$\lvert F\rvert = \lvert C\rvert$	$\lvert W\rvert = \lvert W_F\rvert$
032	12	12	12
034	640	208	208
035	40	14	14
036	240	60	58
037	960	120	116
038	1920	992	992
057	162	132	132
093	360	232	232

from every helper-true state. These checks concern saved strict arenas, independently of the native raw four-fixed-point decision or cooperation of an extracted graph. The 14-input cohort is disjoint from the ten condition-violating models; its correspondence and preservation results do not transfer to them.

Direct and decomposed calculations share the GR(1)/BDD engine. Finite-set qualification on 902 synthetic games covers non-Boolean types, multiple justices and environmental dead ends; it supplies a second representation, not an independently compiled public model.

A separate condition-violating input We also checked Humanoid 044, selected before its released-model run as the smallest of the three previously observed initial flips. This outcome-informed case is separate from the fixed 14-input cohort. The full typed mapping retains both past-time auxiliaries and verifies the original relations against the earlier model. The released SYS relation again retains constraints at $b = 1$, yet $U = D$ and $F = C$. Direct solution and the generalized decomposition agree: $W_F = W_R(\cdot, 0) = \varnothing$ and $W_R(\cdot, 1) = D$. Thus all 2,387 original winning states are lost, including 1,211 through the invariant restriction beyond the immediate loss. The original initial decision changes from true to false; all eight allowed input choices lack a winning initialized response. Its original nonblocking region V was already empty, so this is not rejection of a cooperatively feasible contract. This model-only experiment makes no claim about a native four-fixed-point verdict or an extracted controller.

E. Public source inventory and a cooperative continuation witness

Inventory and certified cohort. The archive contains 449 source paths and 425 byte-distinct units: 423 non-library sources and two pattern libraries. Extraction used representative archive-path order, without prioritizing known outcome labels, with a fresh JVM per input and limits of 60 seconds, 512 MiB heap, and 128 MiB direct memory. After excessive debug output limited the first run, a declared rerun used reversible output compression with the same input order and computation limits. A BDD-file cap followed by a process-cleanup permission failure caused an integrity stop. Its saved outcomes are 70 completed original-model runs, 107 resource UNKNOWNs, 54 technical UNKNOWNs, one integrity UNKNOWN, and 193 unattempted inputs. These are not a completed evaluation of all 425. Of the 70 completed runs, the independent checker certified 59; two exceeded its 64-variable limit and nine failed public-field roster integrity checks involving array metadata. All 59 normally terminated PASS certificates enter the preservation cohort, without filtering by

region sizes or outcomes. The checker shares the compiler-exported predicates and typed encoding; its independent ROBDD computation does not validate the compiler. The C59 reader-index entry (Supplement K) links the complete roster, the extraction receipt, the independent checker's protocol and frozen inputs, and its certification and stop records. These are saved records of the stopped run, not evidence of a completed whole-corpus rerun.

Selection and scope. Unit 188 is the unchanged public source `AutonomousVehicle_sprint02_ver1.spectra` from the Kind artifact.[6] It is outside the fixed 59-model certified cohort. An initial exploratory lexical screen examined the 366 units outside the certified cohort, including both libraries and the eleven uncertified completed models. Eleven units were read in detail, and the retained witnesses were selected by their outcomes. These observations support no prevalence estimate. The source has ten Boolean environment variables, system lane $\ell \in \{1, 2, 3\}$ and speed $v \in \{0, \ldots, 10\}$, no explicit initializer, and no pattern imports or logical auxiliaries. Its typed carrier has $2^{10} \cdot 3 \cdot 11 = 33792$ states.

Source contract. Write o_i, p_i, g_i for obstacle, pedestrian, and green-light inputs in lane i, and let $f_i = g_i \wedge \neg p_i \wedge \neg o_i$ mean that lane i is free. The environment safety relation is

$$(o_1 \Rightarrow \neg o_2') \wedge (o_2 \Rightarrow (\neg o_1' \vee \neg o_3')) \wedge (o_3 \Rightarrow \neg o_2'),$$

and its justice assumptions are $\mathsf{GF} f_i$ for all three lanes. The fourteen system safety clauses require stopping for a red light or pedestrian in the occupied lane, positive speed in a free occupied lane, speed below five at a crosswalk, no direct jump between the outer lanes, and the following responses to current obstacles. A blocked outer lane forces the next lane to be the middle lane. From a blocked middle lane, a next obstacle on side 1 forces side 3, a next obstacle on side 3 forces side 1, and two clear sides require an outer lane. The system justice is $\mathsf{GF} \bigvee_i (\ell = i \wedge f_i)$.

Let s have $\ell = 2$, $v = 0$, $o_2 = 1$, $o_1 = o_3 = 0$, all lights green, no pedestrians, and no crosswalk. This state satisfies the current-state speed duties. The typed next input with $o_1' = o_3' = 1$ is excluded by environment safety but requires both $\ell' = 1$ and $\ell' = 3$ under system safety. Thus no typed system response exists and $s \notin C$. This obstruction uses only the next-lane clauses and is independent of the timing convention for unprimed invariants. The policy and initializer checks below use the saved translation, in which the seven speed duties are current-state obligations.

A sufficient invariant and policy. Put $f = \bigvee_i (\ell = i \wedge f_i)$ and define

$$R = (f \wedge v = 1) \vee (\neg f \wedge v = 0), \qquad P = R \wedge (\ell \neq 2 \vee \neg o_2).$$

The policy selects the next lane by these ordered rules:

1. From a blocked middle lane, choose side 1 if $o_1' = 0$ and side 3 otherwise. Environment safety makes the chosen side clear.

2. From a blocked outer lane, choose the middle lane. Environment safety makes the next middle lane clear.

3. Otherwise, if $o_2' = 0$, choose the middle lane.

4. Otherwise remain in the current outer lane; from the middle lane choose side 1.

[6] AI-assisted analysis and drafting: [7, 8].

Set the next speed to one exactly when the selected next lane is free, and to zero otherwise. These rules define one response to every typed input. From R, every original-legal input has a safe response that enters P. The current speed duties hold because the current state is in R, and the next speed rule establishes them at the successor. The lane rules meet all forced moves and exclude outer-to-outer jumps.

On P, an input with f_2' always selects lane 2. The first rule is inactive there; the second and third rules choose lane 2 whenever it is free next. Consequently $\mathsf{GF} f_2$ implies the system justice. The all-clear input, with all lights green and no pedestrians, is environment-legal from every state and may repeat forever. It satisfies all environment justices. The same policy therefore has an assumption-fair continuation after every legal prefix, including prefixes from $s \in R$. This proves legal-prefix cooperation. Here the environment assumptions are entirely system-independent: cooperation follows automatically from strict winning. The example does not demonstrate a separate strategic difficulty in cooperation.

For every state in P, arbitrary typed inputs also admit some safe response: a blocked outer lane moves to the middle, a clear outer lane may stay, and an unblocked middle lane may move to side 1. The speed rule meets the state duties. Hence $P \subseteq C$. The legal policy preserves P, so it wins in the original game restricted by C. For every initial environment value, choose lane 1 with the speed rule. This lies in P and satisfies the original initializer. Thus $I_M(K)$ and $I_M(W_C)$ are both true, although $s \in K \setminus W_C$. Membership in K follows from the exhibited strict cooperative strategies and the main paper's legal-prefix characterization, which uses its round-to-alternating correspondence and the total-arena soundness and completeness theorem of [3]. No complete K, W, or W_C region is needed for this argument.

Reachability under one cooperative winning strategy. The invariant policy itself avoids s. A second strategy uses one private bit, initially armed, and the same initial selector. If its first current state is all-clear, lane 1, speed 1, and its first next input equals the environment component of s, it chooses lane 2, speed zero. On every other first input it uses the base policy. It then disarms the bit and uses the base policy forever. The exceptional edge satisfies original environment and system safety. After that edge, the next legal base-policy response enters P. Thus every legal play enters P within two responses, and changing this finite prefix does not change the justice argument. The all-clear continuation also establishes cooperation while the bit is armed. One allowed initial input and next input therefore reach s under an initialized cooperative winning strategy. This is existential reachability under a specified strategy, not forced reachability, a synthesized native controller, or a recorded execution. The bit is strategy memory and is not added to the source model.

Meaning for continuation. The loss concerns re-synthesis that retains the full current state, including lane and speed. Fixing only the environment state allows a different initial system response and need not cause refusal. No source initializer was changed to manufacture an initial counterexample. There is no observed repair transaction or deployment benefit. The original full-region computation remains resource UNKNOWN; a separate targeted certificate must be distinguished from completing that computation.

Targeted saved-model evidence. Two fresh model-only executions export the original and released transformed arenas before solver construction. The original has 33792 typed states and 32 Boolean encoding variables; the transformed arena has 67584 states and 34 encoding variables, adding only the environment helper. Full typed comparisons of the environment and system relations in all four helper quadrants agree exactly with the literal complete-relation equations. Original

initialization and the ordered justice lists also agree, with the unique added $\mathsf{GF}\neg b$ assumption in position four. In particular, the system relation is unrestricted when the current helper value is $b = 1$. This is a separate case from the fixed 14-input comparison and its retained constraints.

On the fresh original predicates, a targeted checker verifies thirty obligations implementing the preceding proof. It establishes $|R| = 3072$, $|P| = 2560$, $|C| = 6656$, and singleton displayed states. The policy covers all $33792 \cdot 1024 = 34603008$ typed current/input pairs with exactly one response each. Safety and invariant entry are checked on every original-legal policy edge from R; universal totality alone is not treated as safety. Ordered justice identities, the all-clear fair continuation, the quantified initializer, the response obstruction at s, and the exceptional initialized edge all pass. A separate synthetic qualification has one positive case and nine deliberate predicate mutations, each producing exactly its expected failed obligations. It does not count as public-model evidence.

The target checker performs no winning-region fixed point or controller extraction. Combining its invariant certificate with the independently audited full helper correspondence gives $s \in K \setminus W_R(\cdot, 0)$ and $I_M(K) = I_M(W_R(\cdot, 0)) = \mathit{true}$. The latter equality concerns strict initial feasibility in the transformed arena, not cooperation there or a native four-fixed-point verdict. The checks retain all original variables and their typed encodings.

Independent source correspondence. A separate audit directly encodes the unchanged source's fourteen system-safety clauses, three environment-safety clauses, and four ordered justice predicates. Since the source has no explicit initializers, both initial predicates are true on the typed domain. Its seven speed constraints remain current-state obligations; no implicit next-speed or initial restriction is added. An independent saved-DAG branch reader agrees on all eleven exported predicates, including the three typed-domain factors. Raw-domain checks include unused lane and speed encodings. The initial and justice predicates agree on all 33792 typed states. Decoded variable support justifies exact quotient comparisons of each safety relation over all $33792^2 = 1141899264$ typed current/next pairs: system safety uses 4460544 representatives of multiplicity 256; environment safety uses 64 of multiplicity 17842176. Unreachable and environment-illegal pairs are included. All differences are zero. Qualification uses 97 hand-derived source cases and 51 artificial boundary checks. This strengthens the source-to-original-predicate correspondence for this input; it proves neither general compiler correctness nor a complete winning region.

F. A residual-only public source witness

Unit 120, `ExitModule_ParkingSpec_194.spectra`, is another outcome-selected source outside the certified cohort.[7] Its witness is source-level only; it has no newly generated or certified model. The environment valuation is front sensor zero, side sensor one, parallel parking status, side status NONE, black color, alignment true, and done false. The system valuation is STOP, STRAIGHT, velocity zero, and PARKING. All ten instantiated response-pattern monitors are at their initial state S_0.

A typed next front-sensor value of one simultaneously activates an exit guarantee requiring next velocity one and a STOP guarantee requiring next velocity at most the current zero. Both constraints refer explicitly to next velocity, so the conflict does not depend on the timing convention for unprimed invariants. The input is environment-illegal: stopping freezes the front sensor at zero. Thus the full residual state lies outside C.

On every legal input, choose PARKED, STOP, STRAIGHT, and velocity zero, then retain those values. Stopping freezes the relevant sensor, alignment, color, and done values; parallel status remains fixed and the unrestricted side-status input has no active effect. All ordinary guarantees hold, and

[7] AI-assisted analysis and drafting: [7, 8].

the system parking justice holds from the first successor. The eight environmental response-pattern triggers require forwarding or reversing, so they remain inactive. In the two system response patterns, respectively, the response is already true at velocity zero or the trigger is false. Each monitor remains at S_0 according to the imported `pRespondsToS` update and meets its $\mathsf{GF}S_0$ justice. The three ordinary environment justices contain the stopping disjunct. Every legal prefix therefore has a fair continuation under this residual strategy, with monitor state retained.

The declared initializer requires alignment false, whereas this witness has alignment true. We establish neither its reachability from the original initializer nor initial cooperative feasibility. The literal semantic theorem gives residual loss for the stated monitor expansion. The compiler's auxiliary ownership and any support generated by the unused past-time definition require separate checks before this reasoning transfers to a saved released model. This source witness is not included in any computational cohort or count.

G. A published learned repair tested for cooperative admission

We additionally test the preferred learned Arbiter solution in Appendix C.4.1 of [2].[8] The same declared reconstruction from the eleven-candidate cohort restores the omitted response-pattern declaration: 886 bytes, no imports. After inspecting prior results, we selected the smallest candidate with empty original V, K among those previously positive under residual initialization. This outcome-informed case is separate from the fixed 14-input public-corpus comparison. Cooperative admission adds to the degradation study's ordinary-realizability objective, which we do not dispute or historically reproduce.

Source obstruction. The environment fields are a, r_1, r_2, the public system fields are g_1, g_2, and two system-owned response monitors are retained. The original environment safety is

$$E(s) = a \vee (\neg g_1 \wedge \neg g_2 \wedge \neg r_1 \wedge \neg r_2), \qquad \rho_e(s, e') = E(s), \quad \theta_e = \mathit{true}.$$

Because this current-only invariant refers to system variables, the audited StateInvTranslator leaves it as current safety; it neither strengthens the environment initializer nor globally primes the formula. Ordinary system invariants contribute current initialization and next-state safety. The response monitors update from current requests and grants.

If $E(s)$ is false, no legal environment input exists. Otherwise every next input is legal, including $(a', r_1', r_2') = (0, 1, 0)$. If no safe response exists, strict winning fails. Every safe response instead reaches t with $E(t) = \mathit{false}$, regardless of system values, auxiliary states, or private memory. That legal prefix has no infinite legal continuation, hence no assumption-fair continuation. Thus no residual state admits a strict cooperative strategy. The nonempty original initializer then excludes initial cooperative realizability; this does not assert failure for artificial vacuous initializers.

Complete output and independent check. The configured released job returns a complete 112-state, 1,176-edge native graph with eight initial states covering all initial environment choices. The reachable helper-false view contains 56 states and 280 edges, preserving native IDs, rank memory, both monitors, and initial choices. Its original safety and input coverage pass; strict liveness holds, but 21 reachable states have no assumption-fair continuation. An independent implementation directly evaluates the saved decision DAGs against source-coded equations on all 128 original states and 16,384 state pairs. It checks the native printer, ordered justices, 1,792 native input candidates

[8] AI-assisted analysis and drafting: [7, 8].

and 448 original-view input candidates. Bitset transitive closure independently establishes the graph properties. It imports no production BDD or fairness checker; its result was sealed before comparison with that checker, after which the state, edge, justice and SCC sets agreed. Both checks share compiler exports; this source-equation check is specific to the contract, not a general compiler proof.

Two attempts and limits. The first native attempt stopped at an 8-MiB stdout cap and remains UNKNOWN. A separately recorded followup kept the source, runtime, classes, synthesis settings, extractor, printer and checker unchanged. Native stream caps rose from 8 to 64 MiB, the wall bound fell from 60 to 10 seconds, and the output postcheck rose from 64 to 128 MiB; checker limits stayed fixed. It completed with all 11,927,997 stdout bytes retained, normal teardown and child reaping: one complete result from two attempts on one candidate. No transformed strict-winning region was calculated. Neither native acceptance nor the complete-literal $C = D$ result establishes preservation in this released arena. No repair learning, adaptation loop or deployment was executed; other cohort counts remain unchanged.

H. The fixed eleven repair candidates and selected Lift behavior

The completed comparison covers all eleven replacements in Appendix C.3–C.4 of [2]: five preferred learned candidates and six trivial alternatives from five examples.[9] Two Minepump alternatives use the unchanged listings. The other nine use the declared executable reconstructions appending the paper's omitted response-pattern declaration; Traffic Single P also lowers its direct eventuality to that pattern. Lift's one-floor assumptions and movement guarantees are unchanged from the published replacement listings. Its appended pattern includes auxiliary initialization, deterministic updates, and justice [2, Appendix C.2.1, C.3.3, and C.4.3]. These are correlated published candidates, not eleven independent applications. The constructed controls in Section I are excluded. The extraction intervention and modeled episode below are later followups on existing inputs; they do not alter this cohort's released-route outcomes or denominator.

Traffic Single's direct-eventuality lowering. The learned Traffic Single guarantee $\mathsf{G}(s \Rightarrow \mathsf{F}p)$, with $s = \mathit{car}$ and $p = \mathit{green}$, uses the paper's response pattern. Writing $q = 0$ for $S0$ and $q = 1$ for $S1$, its deterministic monitor is $q_0 = 0$, $q' = (q \vee s) \wedge \neg p$, with justice $\mathsf{GF}\neg q$. Every infinite observable word has exactly one monitor run. If every trigger receives a current or later response, each pending obligation clears; either clearances recur infinitely or q eventually stays zero. Conversely, an unanswered trigger makes q permanently true from the next step, violating the justice. Hence the two forms have the same infinite observable trace language after projecting out q. This is the saved reconstruction argument; compiler correspondence, auxiliary ownership, and Mealy support remain separate obligations. The guarantee retains SYS ownership. The equivalence starts at the original initializer; continuation checks retain the reached monitor value.

Initialization, correspondence, and output checks. For contract-feasibility checks, every case uses its original initializer, including all logical auxiliaries. For the independently certified original cooperative region K, we check

$$I_M(K) = \forall e : \theta_e(e) \Rightarrow \exists y : \theta_s(e, y) \wedge K(e, y),$$

where y includes the logical system monitors. All eleven have nonempty legal initial environment domains; eight pass this predicate. The saved region authority uses the original nonblocking arena

[9] AI-assisted analysis and drafting: [7, 8].

and the normalized round/alternating correspondence of Section A. Neither a residual trace entry nor $K \neq \varnothing$ substitutes for initialization.

Minepump P: strict initialization and the native refusal. The saved original strict region contains 28 of the 32 typed states. Its three-node Boolean representation is

$$W = \neg \mathit{highwater} \vee \neg \mathit{methane} \vee \mathit{pump}.$$

The original environment initializer fixes highwater and methane to false; the system initializer fixes pump to false, the response monitor to $S0$, and the previous-pump auxiliary to false. Substitution into the saved W gives $I_M(W) = \mathit{true}$, while the existing nonblocking and cooperative certificates give $V = K = \varnothing$ and $I_M(K) = \mathit{false}$. This is a static reading of the saved region and initial predicates, not a new solver run. The historical residual trace entry has highwater and methane true and pump false, so it lies outside W; it is not an original initial state. The configured route's native False cannot be transferred as a claim of original strict unrealizability. Cooperative rejection instead follows from the empty V; no complete current released strict-arena explanation of that raw refusal is asserted.

The released route and settings are those in the main paper. Complete native outputs are preserved in the study records. The Lift static addendum omits Lift P's historical 220,856,594-byte stdout, so that stream cannot be independently rehashed from the addendum. The correspondence-gated audit requires typed equality of initial, safety, auxiliary-update and ordered-justice predicates under identity or the helper-false mapping. The environment initializer is also checked over the full native typed domain. For the nine newly executed candidates, separate implementations check admitted graphs for original initial inputs, every legal next input, the selected safe response and private-memory successor, then strictness and cooperation. Saved comparisons require equal state/start/edge sets, ordered justice labels, SCC evidence and retained raw nodes, including ranks and auxiliaries. A failed correspondence or incomplete output remains UNKNOWN at this gate.

Independent hand-derived source equations are also compared with saved initialization, safety, and ordered justice predicates over the full typed domains, retaining logical auxiliaries. For Arbiter P and Traffic S P, the prior audits use the helper-false view of the released model; a separate nine-candidate batch checks helper-free originals for every remaining candidate. These comparisons use the documented translation conventions. Source-to-original equality does not resolve Lift P's distinct released-model mismatch below.

Attempt history and denominator. Arbiter P reuses the complete followup in Section G; Traffic S P reuses its earlier 52-state/496-edge native baseline. Exact certificates and source-byte mappings are retained. Each remaining candidate received one declared native attempt. Six returned graphs passing both checks, Minepump P returned native false, and both Lift attempts stopped at the 64-MiB stdout cap, not the ten-second wall bound. Those two attempts remain resource UNKNOWN.

One separately authorized additional attempt per Lift candidate kept the sources, runtime, synthesis settings, extractor and audits fixed. Stream caps rose to 256 MiB and the per-case output threshold from 128 to 512 MiB; the ten-second native wall and 1-GiB RSS limits were unchanged. Both completed with full stdout: 220,856,594 bytes for Lift P and 219,540,020 for Lift T. The former still fails original correspondence; the latter has a certified strict-only view. At this gate, ten candidates return native graphs, nine have complete original-view certificates, seven pass both properties, two pass strictness only, and one returns native false. Followups do not add candidates or rewrite prior attempts.

Table 6: Correspondence-gated audit of the fixed eleven candidates. P denotes preferred/learned, T trivial, and S/U Traffic Single/Updated. Native False is the solver's negative verdict; dashes mean no controller. Question marks mean original correspondence failed. ^{b} denotes a reused prior baseline and ^{f} a separate Lift resource followup. This table retains the released-route baseline; the later extraction intervention and the post-hoc direct Lift P audit below are separate.

Candidate	$I_M(K)$	Native	View states/edges	Strict	Coop.
Arbiter P^{b}	F	Graph	56/280	Pass	Fail
Arbiter T	T	Graph	56/448	Pass	Pass
Lift P^{f}	F	Graph	?	?	?
Lift T^{f}	T	Graph	154/1170	Pass	Fail
Minepump P	F	False	—	—	—
Minepump T1	T	Graph	12/40	Pass	Pass
Minepump T2	T	Graph	4/16	Pass	Pass
Traffic S P^{b}	T	Graph	16/80	Pass	Pass
Traffic S T	T	Graph	12/80	Pass	Pass
Traffic U P	T	Graph	68/323	Pass	Pass
Traffic U T	T	Graph	56/416	Pass	Pass

A separate original-initializer synthesis route. After recording the historical outcomes above, we froze the same eleven source files, their order, and the settings for one additional route. It applies original-V restriction and justice normalization $G \mapsto G \wedge A$, then the existing normalized four-fixed-point solver and unchanged generic graph constructor. It adds no helper and does not apply the final-choice preference used in the Lift intervention. The source initializers remain unchanged: every legal initial ENV choice must have a SYS/AUX response satisfying $\theta_s \wedge K$, and extracted controllers initialize their private rank to zero. Saved replay or residual entries are not substituted for these initial conditions. All eleven source ENV and joint initializers are nonempty.

Each source received one serial native process attempt, with concurrency one and zero retries. Eleven attempts led to eight solver/extractor calls; Arbiter P, Lift P, and Minepump P stopped before those calls on independently confirmed empty V. All eleven original arenas, including initializers, safety, typed auxiliaries, and ordered justices, match their source-bound saved comparators. The full V and K Boolean functions also agree, including all AUX valuations. The comparisons use exact exported predicates and shared parser-translated BDDs; they are not an independent compilation.

All eight complete graphs pass the original-contract checks for initial coverage, legal-input coverage, safe responses, private-memory successors, strictness, and legal-prefix cooperation. The three refusals have false $I_M(K)$ and no controller verdict. This run has no UNKNOWN or unattempted row. Its graph certificates come from one original-contract checking implementation; the two-judge Lift experiment and the separate fourteen-input comparison retain their own scopes.

Interpretation and limits. The historical table includes replay/residual-entry provenance, so the new eight accepted graphs and the historical seven passing outputs are not a paired success-rate comparison. Historical refusals, resource UNKNOWNs, and Lift P's released-model correspondence failure remain unchanged. For Lift T, the new certificate records zero copied source ENV justices and a 154-state, 1,176-edge graph with two initial nodes. Its equal size does not establish identity with the intervention graph; no graph-equivalence comparison was performed. The successful route supplies an existing alternative without the new preference, while the paired same-game, same-memory intervention still shows one sufficient local change. This whole-route comparison does not isolate the effects of V restriction, normalization, helper removal, initialization, or extraction, and establishes

Table 7: Separate normalized-route results from the original initializer, in the fixed execution order. Init. counts graph initial nodes. Coop. is legal-prefix cooperation. Empty V is an independently checked refusal before solver and extraction; its controller flags are null, shown as dashes. This table does not replace Table 6.

Candidate	$I_M(K)$	Outcome	States/edges	Init.	Strict	Coop.
Arbiter P	F	Empty V	—	—	—	—
Arbiter T	T	Graph	56/448	8	Pass	Pass
Lift P	F	Empty V	—	—	—	—
Lift T	T	Graph	154/1176	2	Pass	Pass
Minepump P	F	Empty V	—	—	—	—
Minepump T1	T	Graph	12/40	1	Pass	Pass
Minepump T2	T	Graph	4/16	1	Pass	Pass
Traffic S P	T	Graph	28/152	8	Pass	Pass
Traffic S T	T	Graph	20/120	8	Pass	Pass
Traffic U P	T	Graph	116/546	8	Pass	Pass
Traffic U T	T	Graph	84/640	8	Pass	Pass

neither preference necessity nor a general synthesis failure. It does not test Friendly's specialized strategy construction or add an eleven-case Slugs run. No new route output was installed or executed in the maintenance-to-service episode. The fixed correlated sources and single attempts establish no timing superiority or broader generality.

Source clauses shared by Lift T and Lift P. The Boolean environment inputs are b_1, b_2, b_3, c and the system outputs are f_1, f_2, f_3. Write m_i for the SYS-owned logical response monitor of button i, encoding $S0 = 0$ and $S1 = 1$. Let $\mathrm{One}(f)$ mean exactly one true floor. The common environment safety relation is

$$E_L = \mathrm{One}(f) \wedge \bigwedge_{i=1}^{3} \left[(b_i \wedge f_i \Rightarrow \neg b_i') \wedge (b_i \wedge \neg f_i \wedge \neg c \Rightarrow b_i') \right].$$

One-floor consistency is thus a condition on the *current* outputs in environment safety. The movement guarantees are $\bigwedge_i (f_i \Rightarrow D_i)$, where

$$\begin{aligned} D_1 &= (f_1' \wedge \neg b_2 \wedge \neg b_3) \vee (f_2' \wedge b_2) \vee (f_2' \wedge b_3) \vee (f_1' \wedge c), \\ D_2 &= (f_1' \wedge b_1) \vee (f_2' \wedge \neg b_1 \wedge \neg b_3) \vee (f_3' \wedge b_3) \vee (f_2' \wedge c), \\ D_3 &= (f_3' \wedge \neg b_1 \wedge \neg b_2) \vee (f_2' \wedge b_1) \vee (f_2' \wedge b_2) \vee (f_3' \wedge c). \end{aligned}$$

Every button and cancellation guard here uses the current value. System safety conjoins these movement implications with

$$\bigwedge_{i=1}^{3} \left[m_i' \leftrightarrow (m_i \vee b_i) \wedge \neg (f_i \vee c) \right].$$

In particular, it does not add $\mathrm{One}(f')$: a movement disjunct can permit extra true next floors. Initialization sets $b = 000$, $f = 100$, and $m = 000$, leaving c unrestricted. The environment justice list is empty; the three system justices are $\mathbf{GF}\neg m_i$. Lift T has $\rho_e = E_L$. These are the clauses of `lift__trivial_1.spectra`, lines 3–81, shared with `lift__preferred.spectra` except for the latter's two added assumptions stated below. Supplement K's Lift T entry links the source, policy, positive graph, complete extension, and selected graph. The policy's private priority bit is separate from all three logical monitors.

Lift T: feasibility and a failing selected graph. The typed original carrier has 1,024 states, including three response monitors $m_i \in \{S0, S1\}$. The saved regions have $|V| = 384$ and $|K| = 316$. Both legal initial choices, distinguished by cancellation input c, have the unique initialized system/monitor response in K. The native graph has 794 states and 13,874 edges; its original view has 154 states, 1,170 edges and two initial nodes. Both checkers establish strictness and reject cooperation.

Table 8 records the original-legal prefix $0 \to 10 \to 339 \to 438 \to 704 \to 680$. It first leaves both K and V on its last edge. All five prefix edges satisfy original safety and the exact current-state monitor updates. The final move is allowed by the next-f_3 disjunct of the current floor-2 move constraint. Its resulting two true floors violate the environment's current-state one-floor assumptions, making every next original environment input illegal. Nodes 680 and 708 are the only two original-view nodes outside both K and V; each has zero original-view successors and 16 native successors, all helper-true. The original environment justice list is empty, but a finite dead end still fails the infinite-extension requirement.

Table 8: Recorded original-legal Lift T prefix, starting at native initial node 0. Bit strings use index order 1,2,3, with $S0 = 0, S1 = 1$; the helper is false throughout. The recorded native system-justice index r names $\mathbf{GF}\neg m_{r+1}$, not a decreasing proof rank.

Node	$b_1b_2b_3$	c	$f_1f_2f_3$	$m_1m_2m_3$	r
0	000	0	100	000	0
10	010	1	100	000	0
339	001	1	010	000	0
438	100	0	001	000	1
704	101	0	010	100	2
680	101	0	011	101	0

For the same observed next environment tuple and the same required next monitor tuple $(S1, S0, S1)$, changing only next f_2 to false gives $(f_1', f_2', f_3') = (0, 0, 1)$. This satisfies source safety and lies in saved $K \cap V$. It is not a selected successor of node 704, and we assign it no replacement native rank or private memory. This one-step alternative is not a repaired controller or a composed strategy. It is also distinct from the later extraction experiment: at first creation of a bad node, the final native candidate retains floor 100, whereas the nearest-Y progress filter in rank selection has already removed 001. On the displayed $704 \to 680$ edge, reuse of an existing node has already removed every continuable candidate. The diagnosis below distinguishes these stages.

A separate positive source policy. A fixed, independently checked policy directly witnesses original initial feasibility. It always sets exactly one floor true and uses one private priority bit $d \in \{1, 3\}$, initially 3. From floor 1 or 3 it moves to floor 2 exactly when another floor is requested. At floor 2 it chooses a requested endpoint, stays if neither is requested, and uses d if both are requested; a departure toward j sets $d' = 4 - j$. Decisions use current buttons. Encoding $S1$ as true, every logical monitor updates as $m_i' = (m_i \vee b_i) \wedge \neg(f_i \vee c)$.

Every move satisfies an explicit source disjunct. The one-floor invariant keeps legal environment successors available. Take a state at time t with b_i true. If f_i or c already holds, the request is answered. Otherwise its button persists until arrival or cancellation. An endpoint request can be delayed by at most one trip to the opposite endpoint before priority reverses, so the requested floor is reached within four transitions from t unless cancellation occurs first. Arrival or cancellation makes its monitor $S0$ in the next state, because its update reads the current f_i and c. All three

monitors therefore visit $S0$ infinitely often. Exhaustive construction from both original initial states (both values of c, with all three source monitors initially $S0$) produces 128 nodes and 1,020 edges, preserving every legal input, all ten original fields and the private bit. Direct source and independent finite-graph checks certify strictness and cooperation. This separate source-coded existence witness uses no native graph, rank, saved K/V restriction or repair as a construction input; it makes no claim from arbitrary orphan monitor states or every state of K.

Extension to the complete transformed model. The same positive policy also extends to a strict strategy for the complete saved transformed model, with environment helper h. Keeping its exact source AUX updates and private priority, we checked all 49,152 prescribed transitions over every one-hot current floor, public input, AUX valuation, priority and current/next helper value. Every response satisfies both original system safety and complete native system safety. This includes all first-flip and post-flip inputs. On this one-hot carrier, which the policy preserves, the helper cannot revert on a native-legal edge. Initialization, domains and ordered justices were checked on all 2,048 native typed states. We do not approximate native system safety by merely disjoining the helper into the source relation, because AUX updates remain required.

Starting from either original initial input, a play that keeps the helper false is an execution of the certified 128-state source controller. If the helper rises, it remains true, falsifying the environment justice $\mathbf{GF}\neg h$, while system safety continues. Thus the policy is native-strict and its helper-false view is cooperative. This establishes neither liveness from arbitrary orphan-AUX states nor cooperation after a helper flip. This source-policy argument alone does not attribute the selected controller's failure; the following paired experiment supplies separate evidence about extraction choices.

Lift T: controlled extraction and modeled replacement

For this Lift T instance, changing one extraction preference while holding the solved game and fixed-point memory fixed produces a strict, cooperative original-contract controller. A subsequent constructed execution uses the checked controller after an environment-model weakening. Both are followups designed after inspecting the baseline failure, not held-out validation or a general synthesis correction. [10]

Baseline and fixed-point agreement. A fresh run reproduces all 23 saved semantic artifacts of the released baseline, including its complete 794-state, 13,874-edge native graph and 154-state, 1,170-edge original-contract view. Both graph auditors again accept strictness and reject cooperation. A behavior-preserving observer uses the same model and memory objects; complete graph equality, semantic equality of the saved model and memory predicates, and unchanged variable order establish observation parity.

A separate finite-set implementation evaluates the inspected raw four-fixed-point recurrence on all 2,048 transformed typed states. It uses the saved transition predicates and their original move quantifiers, without an *Alive* restriction or justice normalization. Its region contains all 2,048 states, exactly matching the native region. The effective 12 X slices, 12 Y slices, three Z components and final winning set also agree on all 57,344 state memberships. The recurrence uses Python integer sets; a shared saved-BDD decoder only reads predicate membership. These checks establish numerical agreement on this input, not compiler independence, the raw core's semantic soundness premise, or a general progress-memory theorem.

[10] AI-assisted implementation, analysis, and drafting: OpenAI Codex; see the main paper's AI-assistance statement.

The transformed strict game permits the dead end. Node 680 belongs to the raw region and satisfies the original response condition C. A focused check of its saved residual strategy also witnesses membership in the actual transformed strict-winning region $W_R(\cdot, 0)$. All 16 legal next inputs flip the helper; none can keep it false. The residual graph has 321 states and 5,136 edges. At every reached node it covers exactly every legal input with a unique safe response, retaining all monitors and private memory. Every successor has the helper true and violates the added environment justice, so every infinite legal play satisfies the strict liveness implication. This uses the complete saved relations, rather than assuming that the syntactic arena matches the complete-literal theorem. The check reuses the saved-BDD decoder; it performs no new native generation or region solution. Thus the game itself permits this non-cooperative behavior. The paired intervention below shows that the same upstream data also permit an accepted output; it does not identify extraction as the unique cause.

Where the continuable alternative is lost. Here the original environment has no justice, and $\mathit{Alive}(s) = \exists e' : \rho_e(s, e')$ equals the current one-floor condition. The trace records two first creations of bad nodes and two later reuses. At first creation, rule a's raw successors have six floor patterns, including the live 001 and 100 alternatives. After its nearest-Y choice, the final five-pattern candidate still contains 100 but not 001; `satOne` selects the dead 011. Later rule a prefers the existing dead node before the final choice. Thus the source-level 001 alternative above and the final-stage 100 alternative have different roles. The trace contains no dropped events or predicate references; it does not by itself certify a fair continuation after any counterfactual choice.

Single-axis intervention and complete-controller checks. Immediately before transition `satOne`, the experimental constructor intersects its final candidate with an original-environment continuation preference. It uses this intersection when nonempty, otherwise retaining the native candidate. With helper h, the preference is

$$h(s) \ \vee \ \exists e' \, [D_e(e') \wedge \rho_e^{\text{native}}(s, e')[h = 0, h' = 0]].$$

Without a helper it uses the original environment relation directly. This leaves helper-true candidates available. It changes no source clause, initializer, solver, stored $X/Y/Z$ set, variable order, rank rule, or earlier existing-state preference. There is no hardcoded floor constraint or node ID.

One solver call supplies the same model and memory objects to the released and experimental extractions. Their initial, transition, ordered-justice and effective memory predicates remain semantically equal; the baseline again matches the saved graph. Both complete outputs are checked by the production and separately implemented graph auditors, whose mathematical checks are unchanged and whose metadata gates explicitly identify the nonrelease variant. Table 9 reports the original-contract views. The experimental full native graph has 778 states and 13,624 edges; 726 of its 13,624 final candidates are refined, with no empty-preference fallback. This is one code-level change affecting many choices, not four patched witness edges.

The six controls were declared before their new executions and retain both verdicts with zero refinements and zero fallbacks. They establish preservation of those successes when the preference is dormant, not generality when it is active. For Lift, unchanged game, raw region and memory suffice for this constructor to return a checked cooperative view after changing concrete extraction choices. One-step availability alone is no fairness proof; the positive conclusion comes from complete-controller checks. No sole-defect, missing-W cause, or general synthesis-repair claim follows.

Table 9: Paired extraction results on Lift T and six prespecified successful controls. Each cell gives original-view states/edges and strict/cooperative verdicts; P/F mean pass/fail. Refinements count candidate sets changed by the preference. These are overlapping followups, not additional members of the eleven-candidate cohort.

Input	Released	Experimental	Refinements
Lift T	154/1170, P/F	154/1176, P/P	726
Arbiter T	56/448, P/P	56/448, P/P	0
Minepump T1	12/40, P/P	12/40, P/P	0
Minepump T2	4/16, P/P	4/16, P/P	0
Traffic S T	12/80, P/P	12/80, P/P	0
Traffic U P	68/323, P/P	68/323, P/P	0
Traffic U T	56/416, P/P	56/416, P/P	0

A constructed environment change and exact handoff. The old contract is Lift T with the additional environment restriction that all current and next buttons are false. A source-coded old controller stays at floor 100 and performs the actual response-monitor updates. Its two reachable cancellation states and four edges pass strictness and cooperation under this maintenance contract. Two modeled plants execute four old-controller steps, with next cancellation values $0, 1, 1, 0$. Removing only the maintenance restriction then exposes the original Lift T environment. Both plants pause at the existing state $b = 000, c = 0, f = 100, m = 000$.

After this change, a fresh paired native run generates both candidates, followed by four fresh graph judgments; no cached candidate is installed. The fixed order is released controller, then experimental controller. One admission policy accepts the first strict candidate; the other requires both strictness and cooperation. Each also requires exact agreement between the current full source valuation and an allowed initial native node. Installation preserves every source field and logical monitor and initializes the new controller's private memory to that matching node. The handoff therefore uses an already initial-compatible paused state; it is not arbitrary-state migration or a reset of pending requests.

Executed outcomes. Both arms execute the five inputs of the known failing prefix, in (b, c) notation: $(010, 1), (001, 1), (100, 0), (101, 0), (101, 0)$. The strict-only policy installs the released controller and reaches floor 011 after these five post-change steps. Independent enumeration finds all 16 possible original next inputs illegal. The strict-and-cooperative policy rejects that candidate, installs the experimental one, and reaches floor 100 instead. It then executes the lexicographically smallest legal input at each step, with false ordered before true in (b_1, b_2, b_3, c). Within the declared 32-step continuation bound, four more transitions give native nodes $680 \to 524 \to 544 \to 434 \to 434$. The final self-loop has $b = 000, c = 0, f = 001, m = 000$, meeting all three system justices; there are no original environment justices. Native IDs are local to each graph: experimental node 680 is not the released graph's dead node 680.

A separate no-replacement counterfactual applies $(010, 0)$ twice to the stationary old controller under the weakened contract. The first step is legal; the second proposed response violates the movement guarantee and is rejected. Thus keeping this old controller does not solve the changed model.

The independent episode audit imports neither the episode runner nor its transition implementation. It checks the 47-event log, source semantics, exact handoff, installed-controller hashes and actual native responses, and binds the four fresh full-graph judgments. It performs 24 environment and 24 system source checks, including the rejected counterfactual step, and 42 native-edge/response

checks. All six supervised processes, comprising the outer execution, one native pair and four judges, complete with verified cleanup. The executed loop witnesses one infinite continuation; the separate complete-graph certificates justify the controller properties. This is a constructed in-process episode with a known-failure prefix, not hardware deployment, newly learned repair, arbitrary memory transfer, or a runtime-cost benchmark.

Bounds, attempts, and provenance. Table 10 gives the declared successful-run limits. Stream bounds apply separately to stdout and stderr. RSS caps apply to the owned process groups, not a measured aggregate tree; output bounds are sampled termination thresholds with final checks, not exact quotas. The episode's outer final `wait4` resource value can include waited descendants on this host. Native and checker children retain a 16-MiB per-artifact limit; the outer episode's per-file limit is 1 GiB so it can capture the already bounded native streams. Each launch requires 1 GiB of free disk space.

Table 10: Bounds for the accepted diagnostic and episode runs. The episode includes serial child processes with their own limits.

Process	Timeout (s)	RSS (GiB)	Stream (MiB)	Output (MiB)
Fresh baseline	10	1	256	512
Observer / paired extraction	30	1	512	768
Finite-set / memory comparison	120	1	1	32
Production graph judge	60	1	8	64
Independent graph judge	180	1	8	64
Outer modeled episode	600	2	8	1024

Observed durations of the completed episode. Table 11 extracts recorded `elapsed_seconds` from the successful episode's process receipts. These are supervised wall durations including capture, finalization, and I/O; the timeout column is the declared cap, not an observation. The outer run contains the paired native generation and four serial graph judgments, together with modeled steps and orchestration. Its 28.36 s must not be added to those child durations. The separate saved-event replay occurred later and is outside that outer run. No solver-only or extraction-only time is inferred from the paired process.

Table 11: Recorded process durations for one completed constructed Lift T episode. Nested rows are components of the outer run; these observations are not a latency or scaling benchmark.

Process	Observed wall (s)	Timeout (s)
Outer modeled episode	28.36	600
Paired native generation	10.82	30
Released: production judge	0.12	60
Released: separate judge	8.60	180
Experimental: production judge	0.15	60
Experimental: separate judge	8.35	180
Later saved-event replay	0.11	120

These measurements describe one completed run of the three-floor example after the three retained technical UNKNOWN attempts below. They are not a statistical sample, online adaptation latency, hardware-normalized comparison, worst-case guarantee, or a scaling result beyond the

bounded models. Process-group RSS caps remain limits; they are not measured aggregate-tree memory consumption.

The first observer attempt exceeded its 256-MiB stdout limit and remains UNKNOWN; a separately recorded followup raised only stream and output-tree limits to 512 and 768 MiB. The episode likewise retains three earlier attempts as technical UNKNOWNs. Attempt 01 stopped when the outer single-child supervisor detected a nested driver; the followup called the existing driver mains in the episode process while each native/checker child retained its own supervisor. Attempt 02's stream capture hit the inherited outer 16-MiB file limit; the next attempt raised that outer limit to 1 GiB. Attempt 03 produced the complete event log and five valid inner processes, but its final `wait4` RSS value of 557,793,280 bytes exceeded the outer 512-MiB cap, so it remains resource UNKNOWN. Attempt 04 raised the outer RSS cap to 2 GiB, retained all inner and output limits, and completed; its independent episode audit passes. Scientific inputs, candidate order, handoff, common input prefix and acceptance criteria stayed fixed. Each followup has a separate plan, output directory and receipt; no automatic retry rewrites an attempt.

The author-side records bind source, class, model, graph, checker, process and event-log identities. Their entry points are `CAUSAL_COMPARISON_CLOSURE.json`, `SUCCESS_CONTROLS_RESULT.json`, and `adaptation_episode_04/EXECUTION_RECEIPT.json`, followed by `adaptation_episode_04/independent/AUDIT_RESULT_attempt01.json`. These distinguish the original released result, experimental result and executed episode. The fourth public archive in Section K preserves them with explicit original/copy identities. Its consumer replays saved semantics and the event audit; it does not regenerate the native producer or execute a new adaptation episode.

Lift P: the whole-model timing mismatch. Let $B = b_1 \wedge b_2 \wedge b_3$, $\phi = \neg c \vee \neg B$, and $\psi = f_1 \vee \neg B$. The added source assumptions give the documented original translation

$$\rho_e^P = E_L \wedge \phi(b', c') \wedge \psi(b, f), \qquad \theta_e^P = (\neg b_1 \wedge \neg b_2 \wedge \neg b_3) \wedge \phi(b, c).$$

The mixed invariant ψ is evaluated at the current state; the environment-only ϕ is evaluated initially and at the next input. Both initial cancellation values remain allowed. The reduced helper-false relation instead checks ϕ at the current input. The remaining environment clauses agree with the independently encoded source equations. With both helper values fixed false, all 2,048 assignments to the eleven original bits in the union of environment-relation supports are checked. The relations differ on 32: 24 false-to-true and eight true-to-false.

One witness starts at the original joint initial state $(0,0,0,0;1,0,0;S0,S0,S0)$ and moves to $(1,1,1,1;1,0,0;S0,S0,S0)$ with helper false at both ends. Original environment safety is false, reduced environment safety is true, and both system relations are true. A control with the opposite environment-relation disagreement is also checked. Initializers and ordered justices pass their separate checks. This diagnosis exhausts the environment relation's support, not the full system-relation domain; system safety is checked at the declared witnesses. This blocks transfer of the native verdict at the whole-model correspondence gate. The earlier diagnostic export-schema failure is retained as an integration failure, not a semantic witness. A separate selected-graph audit follows.

Post-hoc direct original-contract graph certificate. We separately audit the same saved native graph using the original predicates, without transferring its native verdict. The view begins at both original initial nodes and retains helper-false responses to every original-legal input, with raw node identities, private ranks, and AUX values unchanged. All 16 input valuations are checked at each of its 136 reachable states. The 958 legal-input responses are unique and satisfy original SYS/AUX safety; no missing response is repaired and no unsafe legal edge is discarded. The resulting

958-edge view passes strictness but fails cooperation. All 16 failing states are environmental dead ends with zero original-legal inputs, rather than missing controller responses.

Hand-derived source predicates with bitset transitive closure agree with a separate original-BDD, value-decoding, and Kosaraju/reverse-reachability route on all saved states, edges, inputs, ordered justices, and SCC evidence. Both routes share the frozen graph and prior source qualification. The BDD route reuses the earlier saved-BDD decoder, and implementation code was visible across routes; this is not independent compilation or a blind implementation comparison. This certificate leaves the whole-model mismatch and the historical correspondence-gated table unchanged. It adds no native synthesis run, new candidate, or initially cooperative false-refusal example.

A source obstruction independent of the selected graph. From either legal initial input, $b = 000$, $f = 100$, and all monitors are $S0$. The environment chooses $(b', c') = (010, 0)$. Source system safety forces $f_1' = 1$. Any additional true floor immediately violates the environment's current one-floor assumption; otherwise the next state has $b = 010$, $c = 0$, $f = 100$, and all monitors $S0$. The environment can then choose $(b', c') = (111, 0)$: button persistence, button clearing, and the next-input invariant all permit this move. System safety forces $f_2' = 1$. An additional true floor violates the one-floor assumption; with only f_2' true, the next state instead violates the mixed current-state assumption $f_1 \vee \neg(b_1 \wedge b_2 \wedge b_3)$. Thus every safe response yields an environmental dead end within two rounds. A strict controller that avoids the first dead end must answer the second legal input safely; hence no strict cooperative strategy exists from either original initial input, regardless of private memory. This written source argument is independent of the selected graph and does not recompute a winning region; it does not claim that the intermediate state lacks an existential continuation. The argument uses the documented translation of mixed ENV/SYS invariants as current-state environment safety. Initially $b = 000$, and both chosen inputs have $c' = 0$, so ϕ is true at every current and next valuation used here; its current/next timing does not affect this obstruction.

I. A declared pair separating region membership from selection

The pair was fixed before native execution and is separate from the eleven published candidates.[11] Choice has one environment Boolean e, one system Boolean p, no logical auxiliary, and

$$\theta_e = \neg e, \quad \theta_s = \neg p, \qquad \rho_e = (e' \Rightarrow p),$$
$$\rho_s = \top, \quad A = \mathbf{GF}e, \quad G = \mathbf{GF}\neg p.$$

Toggle changes only system safety to $p' \leftrightarrow \neg p$.

Original source proof. From every one of the four original typed states, policy $p' = \neg p$ satisfies system safety and G on every play in either contract. After every finite original-legal prefix, the environment can choose $e' = p$ forever; this is legal and makes e alternate. Thus this same policy is strict and cooperative from every state, yielding $W = V = K = C = D$ and nonvacuous $I_M(K) = \mathit{true}$ for both contracts. In Choice, policy $p' = 0$ instead satisfies G itself but forces $e' = 0$ after reaching $p = 0$, so no assumption-fair suffix remains. Every state of that bad policy still belongs to K.

[11] AI-assisted analysis and drafting: [7, 8].

Audited complete transformed relations. For helper h, both saved native models satisfy

$$\begin{aligned} \theta_e^R &= \neg e \wedge \neg h, & \theta_s^R &= \neg p, \\ \rho_e^R &= ((e' \Rightarrow p) \vee h') \wedge (\neg h \vee h'), & \rho_s^R &= \rho_s \vee h, \\ A^R &= \mathbf{GF}e \wedge \mathbf{GF}\neg h, & G^R &= G. \end{aligned}$$

An independent saved-DAG check establishes these equations on all eight typed native states and all 64 current/next pairs in each case. Field ownership, domains and initialization are checked on the full typed native model; justice identities and order are checked at the original, preconstructor and solver phases. Hence the same p-toggle policy satisfies native system safety and G unconditionally, and its helper-false restriction has the cooperative source behavior just proved. The transformation does not exclude this good original-view policy. This does not demand cooperation in the full transformed arena after an environment-chosen helper flip.

Actual selected outputs. Choice returns three native states and seven edges. Its original view has only node 0, with $(e, p) = (0, 0)$, rank 0, and a self-loop. It satisfies G and fails cooperation while remaining inside $K = V = D$. Toggle returns seven native states and 18 edges. Its original view has nodes $0 = (0, 0)$, $1 = (0, 1)$ and $6 = (1, 0)$, all rank 0, with edges $0 \to 1$, $1 \to 0$, $1 \to 6$ and $6 \to 1$. It passes both checks. Each case received one native attempt; complete typed source correspondence and independent graph checks succeeded for both. Saved state, edge, justice, SCC and raw-node comparisons agree exactly.

The pair shows that $I_M(K)$ and membership of every reached state in K do not certify the particular output's cooperation. It does not attribute that selection separately to the raw fixed-point core or the concrete constructor. Both unnormalized original arenas admit legal G-fair, A-unfair plays, so they do not refute the conditional language-inclusion theorem of [3]. The pair adds no public candidate, learned repair, adaptation episode or deployment result.

J. Exact strict-preservation losses in the certified public cohort

We compute C from each of the 59 certified original models by quantifying every typed next system and auxiliary value, followed by every typed next environment value.[12] The quantifier ranges include environment values disallowed by ρ_e. We recheck the saved independent W function and preserve exact typed domains.

Forty-nine inputs satisfy $W \subseteq C$ and ten do not. The latter are one AMBA variant, three Humanoid variants, two PCar variants, one AirportShuttle variant, and three robotic-arm variants. By the main paper's exact preservation corollary, the complete semantic literal helper fails to preserve the entire original strict-winning region on these ten models.

We evaluate all ten condition-violating inputs, without selection by resulting loss or initial decision. Nine exact calculations complete and one reaches the fixed BDD resource limit. Table 12 reports W_C, exact loss, and the original quantified initializer. Three Humanoid variants change from $I_M(W) = \mathit{true}$ to $I_M(W_C) = \mathit{false}$. Their exact losses strictly exceed $W \setminus C$ by 1211, 6112, and 2292 states, respectively. Thus an immediate response-totality check alone misses losses propagated by the invariant requirement. For variant 045, W_C even remains nonempty (eight states), while none of its 20 allowed initial input choices has an initial response in that region. Region nonemptiness therefore does not restore the lost initial decision.

[12] AI-assisted analysis and drafting: [7, 8].

Table 12: Exact loss for all ten condition-violating public inputs. R is existential reachability through original legal joint edges from the original initializer. Every original initial decision is true. The last column reports the literal complete-semantic initial decision $I_M(W_C)$. Dashes retain the one resource-limited calculation; they are not zeros. IDs identify source variants in the fixed artifact roster.

ID	Source family	$\lvert W\rvert$	$\lvert W \setminus C\rvert$	$\lvert W_C\rvert$	$\lvert W \setminus W_C\rvert$	$\lvert R \cap (W \setminus W_C)\rvert$	$I_M(W_C)$
003	AMBA	34575	6144	28431	6144	2048	True
044	Humanoid	2387	1176	0	2387	1581	False
045	Humanoid	11232	5112	8	11224	7294	False
046	Humanoid	42312	37632	2388	39924	2667	False
059	PCar	5196	1600	3596	1600	200	True
060	PCar	5196	1600	3596	1600	200	True
072	AirportShuttle	4601344	4070400	530944	4070400	49352	True
159	RoboticArm	447213	355043	—	—	—	Unknown
178	RoboticArm	892496	361284	531212	361284	24648	True
179	RoboticArm	892496	361284	531212	361284	100140	True

The other six completed inputs retain strict initial feasibility. Every completed loss contains states existentially reachable by original-legal joint execution from the declared initializer. This establishes a possible continuation-state boundary, not reachability under a particular winning controller or an observed runtime execution.

All ten affected original contracts already have $I_M(W_V)$ false, and hence $I_M(K)$ false. Every strict cooperative strategy wins inside V, so this necessary condition excludes cooperative initial realizability independently of the normalized four-fixed-point calculation. We additionally intersect the independently certified K with the exact loss $W \setminus W_C$ for every completed calculation. The intersection is empty in all nine: seven have $K = \varnothing$; the two PCar variants each have 1636 cooperative states, all preserved. Thus these public results demonstrate strict-region loss and three strict initial flips, but no loss of a certified cooperative state within this cohort. They do not demonstrate rejection of an initially cooperative-realizable public contract. Guard establishes that separate possibility. The tenth calculation remains UNKNOWN. Moreover, these counts concern the complete semantic literal equations. They do not by themselves transfer the theorem to the released syntactic route, whose helper-true constraints require separate analysis.

Detailed literal-cohort summary The 59-model cohort demonstrates strict loss under the complete literal equations, but no certified cooperative-state loss or rejection of an initially cooperative contract. Of these models, 49 satisfy $W \subseteq C$ and ten do not. Nine of the ten exact loss calculations complete; one reaches the fixed BDD limit. Three Humanoid variants lose strict initial feasibility, with loss propagating beyond $W \setminus C$. All ten already fail $I_M(W_V)$, excluding cooperative initialization. Every completed loss calculation has $K \cap (W \setminus W_C) = \varnothing$: seven have empty K, and two PCar variants retain their cooperative states.

Detailed released-arena comparison All fourteen comparison inputs satisfy $W \subseteq C$. Thirteen released-arena analyses complete and establish strict-region preservation; one is resource UNKNOWN. Of eight completed helper arenas, Elevator matches both complete guards because original system safety is true. The other seven match the literal environmental relation in every helper quadrant and original system safety at $(b, b') = (0, 0), (0, 1)$, while retaining nontrivial system constraints at $(1, 1)$. These seven reproduce the literal first-flip timing but match neither complete guard. Five

no-helper arenas instead have complete original-model identity. Original initialization and ordered justices are retained.

K. Public evidence and consumer rechecks

The local manuscript source package includes `EVIDENCE_READER_INDEX.txt` and `EVIDENCE_READER_INDEX.json`. The indexes are public at documentation version `residual-census-evidence-20261021-v1`. This version adds the residual-entry static snapshot while retaining six earlier scientific archives, the constructed Guard source, and selected Minepump records [1]. The indexes map sixteen claim groups to archive members: inputs, source listings where provided, saved results, checkers, and expected outcomes.[13] The index adds navigation, not scientific validation or support for every historical rerun. The earlier version `saved-evidence-20261020-v1` supplies three downloads, `SHA256SUMS`, `RELEASE_MANIFEST.json`, and a README with integrity and source-restoration commands. The public archive hashes differ from the historical ZIP hashes: 97 third-party text copies are omitted and retrieved from their original distributors by hash-checked scripts. All retained archive members preserve their bytes from the earlier anonymous evidence packages. These packages distinguish original execution identities from anonymized or compressed copies in their manifests and ledgers. Source restoration is required before checking those packages' complete file trees or using their closed-package recipes. Neither the download check nor restoration runs a scientific calculation; saved results and qualification limits remain unchanged.

Core. `saved-evidence-public-20261020-v1.zip` covers the 59-model preservation analysis, the same 14 inputs across four paths, and the 11 published replacements. Start with `PAPER_EVIDENCE_INDEX.tsv`, `PUBLIC_14X4_OUTCOMES.tsv`, and `READER_GUIDE.txt` inside `seams2027-saved-evidence/`. The index links the groups' saved inputs and results. `CONSUMER_RECIPE.txt` describes one representative Arbiter initialization recheck, not a rerun of every cohort or native synthesis. Historical correspondence-gated outcomes, including Lift P's UNKNOWN, remain unchanged. The Lift T entry separately links its 128-state positive source policy, 49,152-assignment complete extension, and failing selected graph, with their saved checks. It records prior author-side rechecks; that source-policy extension has no new consumer recheck. The C14 entry also locates the unchanged PCar source and saved initializer argument.

The R11 entry also links repository files `evidence/minepump-original-strict-records.txt` and its JSON companion. They provide selected existing full-W equality records, original-file hashes, and bindings to the public model and initial predicates used in Section H's Minepump inference. These excerpts support static inspection; the full historical auditors and their dependency closure are not added, and no new execution is claimed.

The separate Guard entry links the unchanged 115-byte source at `evidence/transition_guard.spectra` to its saved input digest, native refusal, helper-false predicate certificate, and later initialization-correspondence certificate in Core. The source strategy, observed refusal, and correspondence checks have distinct roles (Section L); neither certificate computes a cooperative winning region or identifies the refusal mechanism. The source is an added repository file, not a change to the historical archives or population counts.

The C14 entry links the Slugs run receipt and its per-input certificates, which identify the saved original models and exported inputs. The export certificates record exact BDD round-trip equality for original initialization, safety, and justice predicates, preserving typed variable ownership, original initialization quantifiers, and Mealy move order. This checks the encoding of shared translated

[13] AI-assisted drafting: [7, 8].

models, not the Spectra compiler independently; returned-graph checks and native refusals retain their separate scopes.

Lift causal comparison and episode. The fourth download, `lift-causal-extension-public-20261021-v1.zip`, has root `seams2027-lift-causal-extension/`. The repository's `LIFT_CAUSAL_EXTENSION.md` and `CLAIM_TO_FILE_INDEX.json` give the commands and exact members. First run the outer integrity/extraction tool, then restore the 14 exact upstream-derived sources using its hash-checked helper. Finally, `consumer/replay.py` recomputes the raw finite-set four-fixed-point region and all 28 observed memory sets, runs both mathematical graph checkers on 16 saved arms (the seven diagnostic pairs and the episode's fresh pair), and replays the episode's saved source/edge/event audit. A fresh extraction with Python 3.10.9 on macOS passes with `PATH` and `JAVA_HOME` empty. The consumer requires Python's standard library, not Java. This is saved-input replay, not new synthesis, extraction or episode execution. Same-object observations, process completion and class identities remain archived producer evidence, checked for consistency but not regenerated. Actual producer bytecode and large process streams are omitted with explicit hashes; this fourth archive alone does not provide a qualified native-regeneration entry point. The original/copy ledger records metadata sanitization separately from preserved graph and predicate content. Failed attempts and their result/process summaries remain. The prior three archives retain their own qualification limits.

Fresh public-source producer qualification. The fifth download, `lift-native-regeneration-public-20261021-v1.zip`, has root `seams2027-lift-public-regeneration-kit/`. Start with the repository's `NATIVE_REGENERATION.md`, its manifest and validation record, and the kit's `README.md`. They give integrity, configuration and execution commands, exact runtime identities, limits and expected outcomes. A bounded run of the exact distributed control files fetches the scientific source, checker, profile, and dependency inputs anew from public URLs. Its 165 requests retrieve 113,412,497 bytes. Selective extraction excludes historical models, controllers, and BDD results from the scientific inputs; restoration recovers 14 source files. Compilation of all six custom Java sources produces 19 classes with the expected byte identities, using the fixed order of 164 JARs. Class-origin checks cover 47 probe classes and the 34 custom/released classes observed in the native execution. One solver invocation supplies the same game and memory to the two extractors. Both complete-graph judges again report baseline 154 states/1,170 edges, strict and non-cooperative, and variant 154 states/1,176 edges, strict and cooperative. All 176 supervised processes complete and are reaped; a separate read-only audit checks the sealed inputs, provenance, graph comparisons, and process records. Historical comparison occurs only after the scientific processes finish.

This qualifies the executed local placement on an installed, pinned JDK 17.0.19 and Python executable under macOS. The JDK is not restored from a public runtime archive; Python's standard library, operating system, platform libraries, and certificate store remain host supplied. It is not a fully hermetic or cross-platform reproduction. It does not regenerate the six inactive controls or execute another adaptation episode. An earlier local run and this qualification run have separate receipts. The latter uses a fresh generated configuration; distributed executable controls retain their executed bytes. Later receipt-path anonymization changes only presentation metadata, with separate original/copy identities and no scientific-content changes. The kit itself was copied locally, not fetched from public hosting during qualification. Anonymous qualification records alongside the kit ZIP identify the executed and distributed manifests and controls. Each later placement requires its own configuration, review and execution record.

Focused saved-state certificate. The kit also supplies `focused/check_node680.py` with a separate bounded command. Its executed copy verifies the public saved model and graph, the original/copy ledger and the permitted digest bridge before checking every node's typed valuation, rank and predicate bindings. It checks all 2,048 valuations of the sole environment justice, which is exactly $\neg h$. At saved node 680, all 16 original source inputs have three safe responses, but no original-legal next input remains; all 16 native-legal inputs flip h. The 321-state, 5,136-edge residual covers every legal input safely and keeps h true thereafter. This proves actual transformed strict winning from this full helper-false valuation by falsifying environment justice forever. It neither equates the entire raw core with that winning region nor identifies extraction as the sole cause. The recheck uses the shared public BDD decoder and saved inputs; it performs no compilation, synthesis or new episode execution.

Table 13: Support availability and replay status in the manuscript and saved archives. One item may serve several evidence roles described in the main paper. An index entry does not establish an independent rerun.

Evidence status	What it supports and what remains separate
Written derivations	Proofs and source arguments printed here; cited primary statements still require external comparison.
Shared-export recomputations	Separately implemented region operations on the same compiler-exported predicates; not independent compiler validation.
Author-side saved checks	Recorded region, relation, and complete-graph results; the index locates their inputs and checkers.
Consumer rechecks	One core initialization, three Vehicle scopes, Lift raw/memory, 32 graph judgments and episode-log checks, plus the separate normalized-route checks below; saved-input replay, not native synthesis or a new episode.
Qualified regeneration	Exact distributed controls fetch public scientific inputs, rebuild all custom sources, generate one Lift pair and run four graph checks; same pinned host runtime, not a new episode or control cohort.
Focused certificate	Saved node 680 belongs to transformed strict winning; typed predicate bindings and a complete safe residual are checked, not the entire winning region.
Static inspection only	Lift P's direct graph and source-obstruction addendum; no qualified relocated rerun and no rehash of the omitted historical stdout.

Normalized route on the eleven original contracts. The sixth download, `normalized-eleven-public-20261021-v1.zip`, has root `seams2027-normalized-eleven-evidence/`. Start with the repository's `NORMALIZED_ELEVEN.md`, manifest, validation record and results index. The archive's `README.md` gives exact commands. Its source-only helper reconstructs the eleven omitted paper-derived specifications from versioned publisher HTML and checks the listing and reconstructed bytes; restoration is separate from the consumer. The release qualification uses separately prepared exact sources and does not claim a new download of them.

With those sources in a separate directory, `consumer/check_saved.py` first checks the complete archive manifest, fixed source roster, ten original scientific module identities, and current model, result and controller bindings. It copies the inputs to a fresh work directory and checks all 33 model-to-source bindings. The unchanged graph checker then checks eleven arms: eight complete graphs satisfy strictness and legal-prefix cooperation; three retain empty-V refusals with null controller flags. It also compares 22 original arenas and 22 full typed V/K functions against their saved comparators, and checks eleven lossless stdout captures. Every new semantic-certificate field must agree with its

saved counterpart, apart from six listed top-level provenance fields. The graph checks retain logical AUX, original initialization quantifiers, safety predicates and ordered justices. Each new arm and its two comparators bind the same exact source file.

The bounded wrapper uses Python's standard library and writes separate receipts without replacing historical evidence. Its portable source binder checks exact supplied source bytes and replaces only the original checker's profile-loading and frozen-source bindings. It does not repeat the private raw/diff/profile binding pipeline or independently compile the source semantics. All ten scientific modules retain their bytes; the wrapper adds input binding and orchestration. This is one graph-judge implementation on shared compiler-exported predicates, not two independent graph judges or producer regeneration. No new native synthesis, extraction, compilation or adaptation episode is performed. The original/copy ledger and exact JSON-pointer allowlist separate presentation-path changes and current derived-file identities from unchanged scientific content and historical execution identities.

Residual-entry static snapshot. The seventh download, `residual-census-evidence-20261021-v1.zip`, has root `residual-census-evidence/`. Start with the repository's `RESIDUAL_CENSUS.md` and the archive's `SUMMARY.json`. It contains the eleven exact entries, 22 complete graphs, 44 retained graph certificates, model and memory BDDs, paired diagnostics, and process-completion records for Section M. All pairs are identical and strict and cooperative; the preference makes no refinements among 836 examined candidates. The original initializer verdict remains false. The archive preserves original reference sources and identifies the omitted upstream-derived extractor through its existing source-restoration recipe. It supplies no qualified relocated replay or fresh-host regeneration. Its manifest and complete original/delivered hash ledger distinguish metadata-only location changes from historical execution bytes; an explicit omissions list identifies excluded host and process files.

Vehicle verification. `verification-addendum-public-20261020-v1.zip` separates P1 (policy, initialization, and specified continuation loss), P2 (original/released relations), and P3 (eleven original-source predicates). Entry points under `seams2027-verification-addendum/` are `P1_READER_RECIPE.txt`, `P2_READER_RECIPE.txt`, and `SOURCE_READER_RECIPE.txt`. The index links inputs, expected outputs, and adoption records: three bounded scopes for one Vehicle source; P3's eleven predicates are not the eleven replacement candidates. The original full-region resource UNKNOWN remains.

Lift static evidence. `lift-direct-static-public-20261020-v1.zip` contains the separate direct original-contract graph certificate and source obstruction. Under `seams2027-lift-direct-static-evidence/`, start with `SCOPE.json`, `root/SELECTED_CONTROLLER_ROOT_ADOPTION.json.gz`, and `root/SOURCE_OBSTRUCTION_PROOF_V2.txt`. The index identifies both saved-graph routes and their comparison. This archive supports static inspection, not a qualified relocated scientific rerun; the omitted historical stdout cannot be independently rehashed from it. The direct certificate does not resolve whole-model verdict transfer.

L. Formal details, route configuration, and decision evidence

Sections L.1–L.5 supply the full retained-constraint proof and counterexample, the detailed cooperation characterization, implementation identifiers, mechanism controls, and decision evidence. Section L.6 gives the complete literal-preservation theorem, full-region corollary, and corrected-guard proofs, with the propagation, initialization, and Guard examples. The main paper states and proves

the principal preservation results and specifies their applicability; the detail here supports further verification and the additional examples.[14]

L.1 Retained helper-true constraints

We use the main paper's finite, fully observed typed Mealy setting and strict semantics [9, 4], retaining all logical monitors in $D = E \times Y$. Let W be the original strict-winning continuation region and $C(s) \equiv \forall e' \in E\, \exists y' \in Y : \rho_s(s, e', y')$. For $T \subseteq D$, let M_T retain the original environmental relations and justice objectives while replacing system safety by $T(s) \wedge \rho_s(s, e', y') \wedge T(e', y')$. Write $W_T = T \cap \mathrm{Win}_{\mathrm{str}}(M_T)$; this outer intersection excludes vacuous winners outside T at environmental dead ends. The original-initializer test is

$$I_M(T) \equiv \forall e \in E : \theta_e(e) \Rightarrow \exists y \in Y : \theta_s(e, y) \wedge T(e, y).$$

For relations E_{ij}, S_{ij}, the predecessor below uses $\forall e' \in E : E_{ij}(s, e') \Rightarrow \exists y' \in Y : S_{ij}(s, e', y') \wedge (e', y') \in Z$. The following gives the full retained-constraint hypotheses, statement, proof, and counterexample for the result summarized in the main paper.

Preservation with retained constraints When constraints remain at $b = 1$, preservation depends on both safe entry and continued safety. Let $E_{ij}(s, e')$ and $S_{ij}(s, e', y')$ be the transformed safety relations with current helper i and next helper j. Assume $E_{00} = \rho_e$, $S_{00} = \rho_s$, and $E_{10} = \mathit{false}$. All original typed variables and logical monitors remain in D; the next helper is observed before the system responds. Let A' and G' be generalized Büchi objectives whose corresponding state predicates agree with A and G on $b = 0$ states, respectively. Require $A^R = A' \wedge \mathsf{GF}\neg b$ and $G^R = G'$ globally; the predicates may otherwise be arbitrary on $b = 1$ states. The relations in 01 and 11 may otherwise be arbitrary, with no next-system dependencies in their environmental component.

Let U be the strict safety-winning region for the helper-true subgame, and let F characterize safe entry to that region:

$$U = \nu Z.\ \mathrm{CPre}_{11}(Z), \tag{L.1}$$

$$F = \{s \in D \mid \forall e' \in E :\ E_{01}(s, e') \Rightarrow \exists y' \in Y :\ S_{01}(s, e', y') \wedge (e', y') \in U\}. \tag{L.2}$$

Here CPre_{11} uses E_{11} and S_{11} with the same typed quantifiers and dead-end convention as CPre. Define W_F by replacing C with F in the definition of W_C, including the outer intersection with F.

Proposition L.1 (Preservation with retained helper-true constraints)**.** *Under these hypotheses,*

$$W_R(\cdot, 1) = U, \qquad W_R(\cdot, 0) = W_F.$$

Consequently, full original-region preservation holds exactly when $W \subseteq F$. If the original initializer is preserved and b starts at zero, the transformed initial decision is $I_M(W_F)$.

Proof. Once $b = 1$, every legal continuation keeps it true. The additional environment justice is therefore impossible, so only strict system safety remains; its greatest-fixed-point winning region is U. For any winning strategy starting at $b = 0$, every legal flip must receive a safe response into U. Thus each state reached while $b = 0$ belongs to F. Restricting the strategy to 00 histories preserves the original relations and justice objectives and keeps F invariant, establishing membership in W_F.

[14] AI-assisted analysis and drafting: [7, 8].

Conversely, follow an original winning strategy that keeps F invariant until a flip. The observed flip input has an S_{01}-safe response into U by definition of F. Continue with a safety-winning strategy for U. Such a strategy can be selected uniformly over U in the finite, fully observed safety game. If no flip occurs, the original objective is met; after a flip, system safety holds and the added assumption is false. The preservation equivalence then follows by the residual-strategy argument in Corollary L.3 (Section L.6). □

This is standard safety-game reasoning for a one-way helper, not a new solving algorithm. A nontrivial S_{11} can still give $U = D$; even a proper U can permit every winning active state to enter it. Conversely, the existence of an immediate S_{01} response does not suffice if that response cannot reach U. For example, use one input value, a Boolean response y, true original initial and justice predicates, $\rho_e = \mathit{true}$, and $\rho_s = y'$. Then $W = C = D$. Let $E_{01} = E_{11} = \mathit{true}$, $S_{01} = \rho_s$, and $S_{11} = \neg y \wedge \neg y'$. Every state permits a flip response, but each response reaches $y = 1$, outside $U = \{y = 0\}$. Thus F and the active winning region are empty, despite preserving the original first-flip relation. In the complete literal case $U = D$ and $F = C$; in the complete corrected case $U = F = D$.

L.2 Cooperative existence and selection

Cooperation of a fixed strategy For a strategy f on a total legal arena H, let $L_v(H, f)$ be its infinite plays from v.[15] Legal-prefix cooperation requires

$$\operatorname{Pref} L_v(H, f) = \operatorname{Pref}\big(L_v(H, f) \cap L_v(H, A)\big). \tag{L.3}$$

Every legal prefix must have an assumption-fair suffix under the same strategy and its reached private memory. One fair play is insufficient. Input availability is a separate requirement: the infinite-extension obligation of [4, Definition 3] excludes finite environmental dead ends even with no justices.

Negative conclusions under the all-history reading Definition 3 of [4] quantifies over every environment strategy and every position of its play with the fixed winning system strategy. Restricting those positions to original-legal prefixes gives our requirement. A legal-prefix failure therefore refutes that stronger property for the same original-contract strategy and reached memory. For a complete helper-false view retaining every original-legal input, the witness also survives any completion on illegal inputs; helper-true escapes do not extend that fixed original-contract strategy. This negates one output, not the existence of another strategy. Contract-level negation additionally requires a separate argument excluding every strict cooperative strategy from the original initializer. Conversely, a positive legal-prefix certificate alone does not establish the all-history property.

Applicability of the cooperative core The Kind paper invokes the non-conflicting core's correctness [4, Theorem 3 and Appendix A]. The result of Majumdar et al. [3] has the sufficient premise

$$L_v(H, G) \subseteq L_v(H, A). \tag{L.4}$$

The established normalization $G \mapsto G \wedge A$ ensures this inclusion without changing $A \Rightarrow G$ [3]. Our normalized comparison applies it to the original nonblocking arena, before helper conversion. An invocation of the raw core must separately meet its premise.

[15] AI-assisted text and analysis: [7, 8].

Original-contract comparison regions Strict synthesis is restricted to the nonblocking safety region $V = \nu X.(\mathit{Alive} \wedge \mathrm{CPre}(X))$, where $\mathit{Alive}(s)$ means that some original-legal next environment input exists. This restriction maintains system safety and future input availability against all original-legal inputs; it imposes no environment justice. Normalized non-conflicting synthesis places the environment justices among the system justices before invoking its four-fixed-point core on V. We denote the two regions by W_V and K, respectively; W remains the unrestricted original strict-winning region. Any strict cooperative strategy remains in V: cooperation supplies legal continuations, and strict safety supplies safe responses to every legal input. Its reachable residual states therefore form a post-fixed point of the defining operator. In the total bipartite arena H_V, completed states are $Q_0 = V$; δ_0 retains every original-legal input, and δ_1 retains every original-safe response into V.

Lemma L.1 (Round interpretation)**.** *For nonempty V, put all justice labels of finite total H_V on Q_0 only. Replace the one-edge predecessors in [3, Eq. (15)] by* $\mathrm{CPre} = \forall\delta_0\,\exists\delta_1$ *and* $\mathrm{EPre} = \exists\delta_0\,\exists\delta_1$*, with* $\mathrm{CondPre}(U, X) = \mathrm{EPre}(U) \cap \mathrm{CPre}(U \cup X)$*. Each final round vector component equals the Q_0 restriction of the corresponding alternating component.*

The full proof is in Supplement A. The correspondence identifies final values, not all inner approximants or extracted ranks. Empty justice lists are padded by V in rounds and Q_0 in H_V; if V is empty, both regions are empty. Within V, normalization establishes (L.4), the premise of the non-conflicting soundness statement in [3, Theorem 3]. Lemma L.1 and the union definition in [3, Appendix B.1] connect K to that theorem; its soundness and completeness [3, Theorem 3 and Appendix B.2] identify K as the legal-prefix cooperative region.

Correcting the guard does not ensure cooperation In *the transformed arena*, either complete helper guard lets a strict-winning strategy answer a legal flip. Monotonicity then keeps $b = 1$, releasing system safety but making $\mathsf{GF}\neg b$ impossible. Normalization after helper conversion therefore cannot generally repair the pipeline. This neither refutes cooperation of an original-contract helper-false view nor alone disproves (L.4), which may hold vacuously when the system objective is impossible.

Justice: preservation without cooperation Let environment Boolean e and system Boolean p have true initial predicates and

$$\rho_e = (e' = p), \quad \rho_s = \neg p, \quad A = \mathsf{GF}e, \quad G = \mathsf{GF}\neg p. \tag{L.5}$$

Current-state safety is intentional: strict winners keep $p = 0$, forcing $e' = 0$; at $p = 1$, a legal input exists but safety fails. Thus $W = C = W_C = \{\neg p\}$: preservation holds, but no cooperative strategy exists. The all-zero play satisfies G and violates A, falsifying (L.4). Substitution yields raw region $\{\neg p\}$, or $\neg p \vee b$ under either helper guard, with positive initialization. The supplement gives the recurrence; this does not refute the conditional theorem of [3].

L.3 Implementation identifiers and trace locations

The two gate identifiers are `GR1SeparatedGame` and `isEmptyController`.[16] The latter checks a stored justice-level Z set before generic default-ZYX extraction uses the same model and memory. Section C.4 records the historical environment-only initializer caveat; D the complete saved-relation checks; G–H the published-replacement capture limits, followup histories, and graph evidence.

[16] AI-assisted text and analysis: [7, 8].

Table 14: Constructed and reconstructed mechanism controls. Sizes are states/edges; original-contract checks use the helper-false view. A dash means no returned graph.

Input	Result	Native	View	Strict	Coop.
Guard	False	—	—	—	—
Justice	Graph	4/10	2/2	Pass	Fail
Mixed	Graph	5/16	3/6	Pass	Pass
Traffic P	Graph	52/496	16/80	Pass	Pass
Choice	Graph	3/7	1/1	Pass	Fail
Toggle	Graph	7/18	3/4	Pass	Pass

Section H separately reports a nonrelease extraction variant and a paired comparison sharing one solved model and fixed-point memory; the release identifiers below continue to describe the baseline.

The configured route uses Spectra `1.0.0.202402072123` with JTLV, syntactic Piterman reduction, `optChoice=none`, and `TraceInfo.NONE`. A headless loader replaces IDE resource access; the route excludes the Eclipse GUI and default CUDD backend and differs from the historical C-backed coverage driver [6]. The initial test uses the raw region, and the empty-controller gate rejects only an empty stored justice-level region. Loaded-class hashes match the release archive with the inspected sources; this is not an independent source-to-bytecode proof.

The saved `SynthesisAction` bytecode checks `synthesisPitermanReduction`, then invokes `PitermanSyntacticReduction.doReduction` at offset 487 and `TranslationProvider.translate` at 596. Headless `ReleasedKindJobProducer` follows this order before BDD generation (`SOURCE_ROUTE_TRACE.tsv`), separately from Section B's later-snapshot comparison.

L.4 Mechanism controls

Table 14 compares constructed or reconstructed controls: Guard (Section B), Justice ((L.5)), Mixed, Traffic P, and Choice/Toggle. These are not independent public applications.

For *Choice*, let environment Boolean e and system Boolean p have

$$\theta_e = \neg e, \quad \theta_s = \neg p, \quad \rho_e = (e' \Rightarrow p),$$
$$\rho_s = \top, \quad A = \mathsf{GF}e, \quad G = \mathsf{GF}\neg p.$$

From all four typed states, $p' = \neg p$ guarantees G; after every legal prefix, choosing $e' = p$ supplies an assumption-fair suffix under the same strategy. Thus $W = V = K = C = D$ and $I_M(K) = \mathit{true}$ independently of fixed-point computation. Toggle changes only system safety to $p' = \neg p$, retaining these initializers and justice objectives.

Mixed has environment Boolean e, system Booleans d, y, $\rho_e = \neg d$, $\rho_s = \mathit{true}$, $A = \mathsf{GF}e$, and $G = \mathsf{GF}y$. Its initial predicates are $\theta_e = \neg e$ and $\theta_s = \neg d \wedge \neg y$. Its cooperative strategy can keep $d = 0, y = 1$ while the environment may choose $e = 1$ after any prefix. Traffic P reconstructs a repaired traffic-light contract from [2], using its original initializer rather than a residual entry from repair history.

Guard's source has environment Boolean x, system Boolean p (called y in Section B), trivial initial and justice predicates, and $\rho_e = \rho_s = x'$. The constant strategy $p = 0$ is strict-winning; from every history, choosing $x' = 1$ forever supplies a legal continuation. Thus $W = V = K = D$ and $I_M(K) = \mathit{true}$, independently of the mixed-current-invariant convention. The saved job result binds its input hash to this source and reports native refusal with unchanged initializers, one solver call, and no extraction or printing. The saved helper-false predicate certificate and later

initialization-correspondence certificate pass; neither is a complete transformed-arena analysis. The public reader index links these records separately from the public and replacement populations. The observed refusal therefore conflicts with this source strategy, but its cause remains unresolved and it does not cover the distinct semantic implementation.

Justice's helper-false graph keeps $p = 0$ and then $e = 0$: strictness passes, cooperation fails, and the preceding argument separately excludes a cooperative alternative.

Mixed and Traffic P pass both checks with complete input coverage, despite transformed unfair prefixes. Traffic P's single result also appears in the separate eleven-candidate comparison (Sections G–H).

L.5 Detailed decision evidence

Table 15: Decisions from completed checks or marked written derivations. Passing graph certificates cover legal-prefix cooperation, not all-history kind strategies. Supp. locates supplementary sections.

Case	Evidence under the original contract	Permitted decision
Guard (Supp. B)	An original kind strategy exists; the configured released route refuses; its refusal mechanism remains unestablished.	The refusal cannot justify discarding the original contract.
Humanoid variants (Supp. D, J)	Complete literal relations lose strict initial feasibility (also in released arena 044); cooperative initialization already fails.	Literal refusal alone does not establish original strict infeasibility.
Vehicle (188) (Supp. E)	A constructed winning strategy reaches $s \in K \setminus W_R(\cdot, 0)$; cooperation follows from strict winning. Strict initialization is preserved (saved-arena derivation).	A strict refusal at full state s cannot justify rejecting its original cooperative continuation.
Elevator (032) (Supp. C)	$K \neq \varnothing$, but $I_M(K)$ is false; an allowed initial input forces an environmental dead end.	Reject cooperative initialization. Restricting initialization alone would not repair this returned graph.
PCar (058) (Supp. B, C)	$I_M(W_V)$ is true, $K = \varnothing$, and two strict graphs fail cooperation.	Reject these graphs. A separate source argument excludes cooperative initial realizability.
Arbiter P (Supp. G)	Published learned repair: $K = \varnothing$, yet the released route returns a strict, non-cooperative controller.	Reject cooperative admission. Neither strictness nor vacuous $K \subseteq C$ certifies cooperation.
Lift P (Supp. H)	Whole-model correspondence fails; a separate direct graph audit passes strictness and fails cooperation. A written argument excludes cooperative initialization.	Reject cooperative admission using this original-contract evidence; whole-model verdict transfer remains UNKNOWN.
Lift T (Supp. H)	$I_M(K)$ is true, with a separate checked positive policy; the returned graph reaches an environmental dead end.	Reject this returned graph, while retaining the contract as cooperatively feasible.
Choice / Toggle (Supp. I)	$K = D$ for both; Choice returns a non-cooperative self-loop, while the matched Toggle output passes.	Membership in K, even for every reached state, cannot replace a fixed-strategy cooperation check.
Passing graphs	Complete graphs pass strictness and cooperation, including initialization and input coverage.	The graph witnesses existence within its verified state and private-memory scope.
SmartHome1 (223)	Resource limit in the released run; other paths return passing graphs.	UNKNOWN: no semantic ground for rejecting the contract.

L.6 Complete literal-preservation statements and proofs

This section gives the complete proofs, propagation and initialization examples, and constructed Guard counterexample for the statements in Section V of the main paper. The hypotheses below concern strict games under complete relations; they do not establish the semantics of a raw cooperative-core verdict or certify an extracted controller. Local numbering gives the theorem, corollary, and corrected-guard proposition. Section L.1 covers retained helper-true constraints.

Game and semantic scope Let M be a finite Mealy game with nonempty typed input and response carriers E, Y and state space $D = E \times Y$, including all logical monitors.[17] A round receives e' before choosing y', reaching $s' = (e', y')$. Environmental safety $\rho_e(s, e')$ is independent of the next response; $\rho_s(s, e', y')$ includes all system constraints and monitor updates. Initial predicates are $\theta_e(e)$ and $\theta_s(e, y)$. Under strict semantics, system safety is required on every edge whose environmental safety prefix, including that edge, remains valid. An environmental safety violation releases system safety on that edge and thereafter. On infinite safety-compliant plays, liveness requires $A \Rightarrow G$; failure of A does not excuse earlier required safety. Neither relation need be total. Invalid type encodings are outside D; A, G are generalized Büchi objectives. Strategies retain private memory and give typed responses even to illegal inputs. Define

$$\begin{aligned}\mathrm{CPre}(T) = \{s \in D \mid \forall e' \in E : \rho_e(s, e') \Rightarrow \\ \exists y' \in Y : \rho_s(s, e', y') \wedge (e', y') \in T\}.\end{aligned}$$

An *active* continuation has met environmental initialization, all prior environmental safety, and required system duties. $W \subseteq D$ contains states admitting a strict-winning continuation with some private memory. It includes unreachable typed states and vacuously winning environmental dead ends; initialization applies separately below.

For the helper variants, b denotes a fresh, environment-controlled Boolean observed together with the next original input. Table 1 distinguishes four variants. For retained constraints, U is the helper-true strict safety-winning region, and F contains states from which every legal first flip admits a safe response into U. Seven of the eight completed helper-arena audits have literal first-flip timing but retain system constraints when $b = 1$. Thus a matching first flip does not establish complete literal relations. Supplement B gives the full comparison; D gives relation checks.

The literal reduction Kind writes each converted safety assumption as $\rho_n^e \vee \mathrm{next}(\mathit{envHelp})$ and each safety guarantee as $\rho_n^s \vee \mathit{envHelp}$ [4, Sec. 4.1, p. 4]. Its Theorem 1 states: "Then M is realizable if and only if M' is realizable" [4, p. 5]. Writing the helper as b, conjoining the respective clauses, and including helper persistence gives the following complete relations:

$$\begin{aligned} \theta_e^R &= \theta_e \wedge \neg b, & \theta_s^R &= \theta_s, \\ \rho_e^R &= (\rho_e \vee b') \wedge (\neg b \vee b'), & \rho_s^R &= \rho_s \vee b, \\ A^R &= A \wedge \mathbf{GF}\neg b, & G^R &= G. \end{aligned} \tag{L.6}$$

The guards apply to the *entire* safety relations, including logical auxiliary constraints. This semantic hypothesis must be checked for a syntactic route after all translations.

With $b = b' = 0$, both original safety relations apply; with $b = b' = 1$, both are released. The mismatch occurs on the first flip, $b = 0, b' = 1$: any typed e' is allowed, including one disallowed by

[17] AI assistance: [7, 8].

ρ_e, while ρ_s still applies. Define

$$C(s) \;\equiv\; \forall e' \in E\ \exists y' \in Y : \rho_s(s, e', y'). \tag{L.7}$$

There is no environmental-safety antecedent in C. Passing this one-state test is insufficient: a winning strategy must keep it true at later states, as the propagation example below shows. Let M_C have the original environmental relations and justice objectives, but system safety

$$\rho_s^C(s, e', y') \;\equiv\; C(s) \wedge \rho_s(s, e', y') \wedge C(s'). \tag{L.8}$$

Define $W_C = C \cap \mathrm{Win}_{\mathrm{str}}(M_C)$ independently of initialization. If M_C is initialized, use system initializer $\theta_s \wedge C$. The outer intersection with C excludes outside-C states that win vacuously at environmental dead ends; changing system safety alone does not.

Theorem L.2 (Exact strict-winning continuation region)**.** *Under the stated Mealy, strict, and complete-relation hypotheses, the reduction* (L.6) *has, for every $s \in D$,*

$$W_R(s, 0) = W_C(s), \qquad W_R(s, 1) = \mathit{true}, \tag{L.9}$$

identifying a region with its characteristic predicate.

Proof. At $b = 1$, legal edges retain the helper, system safety is released, and every infinite legal play violates $\mathsf{GF}\neg b$; hence all states win.

For necessity, a winning strategy at $(s, 0)$ must answer every legal first flip $(e', 1)$ with a ρ_s-safe response, so $s \in C$. Restrict it to histories with $b = 0$: these have exactly the original legal inputs and obligations. After each such finite prefix its residual strategy still wins, because required safety holds and Büchi conditions are prefix-independent. Repeating the first-flip argument gives C at every reached successor. Thus the restricted strategy wins M_C and $W_R(\cdot, 0) \subseteq W_C$.

Conversely, follow a winning M_C strategy while $b = 0$ and inputs are original-legal. It maintains C, which supplies a safe response to any first flip. The successor then has $b = 1$ and is winning. Without a flip, original safety and liveness coincide; any transformed-illegal input releases strict duties. □

Corollary L.3 (Full-region preservation)**.** *The literal reduction preserves the entire original strict-winning region at $b = 0$ if and only if $W \subseteq C$.*

Proof. $W_C \subseteq W \cap C$ and Theorem L.2 give necessity. If $W \subseteq C$, every original-legal prefix of an original winning strategy has a winning residual strategy. Every reached state therefore belongs to $W \subseteq C$, so the same strategy satisfies (L.8) and wins M_C. □

Loss can propagate The exact lost region is $W \setminus W_C$, which can strictly contain $W \setminus C$. For example, take Boolean input x, response y, trivial justices, $\rho_e = x'$, and $\rho_s = (\neg y \wedge y') \vee (y \wedge x')$. Here $W = D$ and $C = \{y = 0\}$, but every original-legal response from C leaves C, so $W_C = \varnothing$. An initial test of C alone misses this delayed loss.

Preservation from the original initializer For a continuation region T, write

$$I_M(T) \;\equiv\; \forall e \in E : \theta_e(e) \Rightarrow \exists y \in Y : \theta_s(e, y) \wedge T(e, y). \tag{L.10}$$

Original initial realizability is $I_M(W)$; Theorem L.2 and initial $b = 0$ give transformed realizability $I_M(W_C)$. For an originally realizable contract, preservation requires each allowed initial input to admit a θ_s-compatible response in W_C. Since $W_C \subseteq W$, original unrealizability is preserved. Full-region preservation is sufficient but unnecessary: replacing the first term of the preceding example by $\neg y \wedge \neg y'$ gives $W = D$ and $W_C = C = \{y = 0\}$. With $\theta_s = \neg y$, both initial decisions are true although the regions differ.

An original feasible contract can be rejected The Guard control has Boolean input x, response y, trivial initial and justice predicates, and $\rho_e = \rho_s = x'$. The original game has $W = V = K = D$: a constant response is strict-winning, and from every history the continuation with $x' = 1$ remains available. However, $C = \varnothing$. At the initial helper value $b = 0$, the first input $(x', b') = (0, 1)$ is legal in the transformed game, while $\rho_s \vee b$ is false for every response. Thus the complete Section 4.1 clauses make a realizable specification unrealizable.

The existing first-transition correction When it adds a helper, the released semantic implementation uses the additional first-flip guard $b \vee (\neg b \wedge b')$, equivalent to $b \vee b'$ [10]. Consider the corresponding complete semantic variant R^+, replacing only the system safety of (L.6) by

$$\rho_s^{R^+} = \rho_s \vee b \vee b'. \tag{L.11}$$

This existing variant is not a new synthesis algorithm.

Proposition L.4 (Corrected-guard preservation)**.** *Under the same hypotheses, $W_{R^+}(s, 0) = W(s)$ for every $s \in D$, and R^+ preserves initial realizability.*

Proof. Follow an original winning strategy while $b = 0$ and original safety assumptions hold. The first flip releases system safety *on that edge* through $b' = 1$; every later legal edge is released through $b = 1$, and the added environmental justice is false. Without a flip the original obligations coincide. Conversely, keeping b always zero restricts any transformed winning strategy to an original winning strategy. The unchanged initial predicates together with initial $b = 0$ then give equality in (L.10). □

Counterexample scope and released evidence Guard refutes preservation for the complete printed Section 4.1 relations; Appendix A and Kind's separate well-separation theorem are outside its scope. The complete semantic correction differs from configured syntactic arenas whose constraints can remain after the helper becomes true. Sections D–E and J report the applicable strict-preservation checks.

The configured syntactic route also refuses Guard natively, but its complete saved-arena relation audit is unavailable (Supplement L). A retained first-flip safety obligation could explain a strict-game obstruction, consistently with seven other helper arenas. Their checks do not establish Guard's own predicates, however, and a strict-game obstruction alone would not explain its raw cooperative-core verdict. The native cause remains unresolved.

When syntactic constraints remain Section L.1 gives the full generalized statement and proof. With U and F as defined above, under the stated original-relation, one-way-helper and justice hypotheses, the active strict region is W_F: original strict-winning continuations that maintain F. Full preservation holds exactly when $W \subseteq F$; unchanged initialization gives $I_M(W_F)$. Thus safe entry and continued safety must both be checked. The complete literal case has $U = D, F = C$, and the corrected case has $U = F = D$. A retained partial constraint can invalidate either equality.

Limits of the preservation claim A syntactic route needs complete-relation and helper-true safety checks: translation can leave partial constraints unguarded or change invariant timing. AUX presence alone is inconclusive; a total deterministic update need not obstruct a response. The state retains all logical monitors throughout the residual-strategy arguments. Conditions using W require the original winning region; no low-cost runtime precheck is established.

M. Residual Elevator continuation census

This experiment asks whether the fixed final-choice preference used for Lift changes extraction from exact continuation states of Elevator 032. The source and its earlier cooperation failure were known before selection; the complete eleven-state cohort and preference were frozen before these new outcomes. This is a state census on one previously inspected source, not a held-out contract evaluation.[18]

Source and original feasibility. The unchanged `ElevatorLTL_386_Elevator.spectra` from [6] declares environment variable $f = \mathit{floor} \in \{\mathsf{LOWER}, \mathsf{MIDDLE}, \mathsf{UPPER}, \mathsf{NONE}\}$ and system variable $m = \mathit{elevMot} \in \{\mathsf{FWD}, \mathsf{BWD}, \mathsf{STOP}\}$, with no logical auxiliaries. On these declared domains, its complete clauses are

$$\begin{aligned}
\theta_e &= \mathit{true}, \qquad \theta_s = (m = \mathsf{STOP}),\\
\rho_e &= (m = \mathsf{STOP} \Rightarrow f \neq \mathsf{NONE}), \qquad \rho_s = \mathit{true},\\
A &= \mathit{true},\\
G &= \mathsf{GF}(m = \mathsf{FWD})\\
&\quad \wedge \mathsf{GF}(f = \mathsf{LOWER} \Rightarrow m = \mathsf{FWD})\\
&\quad \wedge \mathsf{GF}(f = \mathsf{UPPER} \Rightarrow m = \mathsf{BWD}).
\end{aligned}$$

The assumption reads current values; it does not constrain f' directly. The three system justices retain their displayed order. The previously certified region is $K = V = D \setminus \{(\mathsf{NONE}, \mathsf{STOP})\}$, where $|D| = 12$. Hence the original initializer verdict remains $I_M(K) = \mathit{false}$: the allowed initial input $f = \mathsf{NONE}$ forces $m = \mathsf{STOP}$, with no legal continuation. The cohort contains all eleven states of K, including three original-initializer-compatible states and eight incompatible states (Table 16).

Exact entries and paired construction. For each frozen state $s = (f_s, m_s)$, only the initial predicates are replaced: $\theta_e^s = (f = f_s) \wedge \neg h$ and $\theta_s^s = (m = m_s)$, where h is the native helper. Every source value is fixed, and new controller private memory starts at rank 0; no old controller memory is inherited. The original initializer is saved separately and is not reapplied. The released syntactic route on Spectra `1.0.0.202402072123`/JTLV is unchanged, without semantic correction or justice normalization. One clean Java compilation is followed by eleven fresh JVM processes, one per entry, each solving once and extracting the released baseline and the fixed preference variant from the same game and fixed-point memory objects. There are 22 returned graphs and no retries or substitutions. Per pair, ten body BDDs and the field/domain/justice structure agree before and after the initializer replacement. Snapshots of 34 model and memory BDDs, including the winning and $X/Y/Z$ sets, agree before either extraction and after each; variable order and memory structure are retained. Objects are not shared across different entries.

Two graph-checking routes. Each returned graph receives two certificates, giving 44 checks. The source-specific route transcribes the clauses above and the helper encoding, uses its own native-BDD text reader, and compares the saved predicates on all 24 typed transformed states and 576 transition pairs. Its separate strongly connected component implementation checks graph fairness. This is an oracle for this pinned source, not a general source compiler. The saved-model route instead uses the independently decoded original BDD model, whole-domain correspondence, and the generic finite-graph fairness checker. It shares the original parser/translation exports and does not independently

[18] AI-assisted analysis and drafting: [7, 8].

establish source compilation. Actual class loading, process completion, and captured-file identities are separate recorded provenance obligations.

Both routes bind the machine graph to the complete native printer output, retain node identities and private rank, and derive the helper-false graph reachable from the exact singleton. They check every original-legal input, unique responses and memory updates, system safety, and strict liveness. No restriction to V or K, edge repair, or state merging is applied. Cooperation requires every reachable node to reach an assumption-fair cycle. With no original environment justice, a cycle is still required: a finite dead end cannot pass.

Table 16: All eleven frozen entries. Raw and original graph sizes are nodes/edges and agree between the two outputs in each pair. "Init." means compatibility with the original source initializer. Checks and refinements count final candidate sets examined by the preference variant.

Entry	Floor	Motor	Init.	Raw	Original	Checks	Refined
s01	LOWER	BWD	no	11/68	6/24	68	0
s02	LOWER	FWD	no	16/96	8/32	96	0
s03	LOWER	STOP	yes	11/68	6/24	68	0
s04	MIDDLE	BWD	no	11/68	6/24	68	0
s05	MIDDLE	FWD	no	16/96	8/32	96	0
s06	MIDDLE	STOP	yes	11/68	6/24	68	0
s07	NONE	BWD	no	11/68	6/24	68	0
s08	NONE	FWD	no	16/96	8/32	96	0
s09	UPPER	BWD	no	11/68	6/24	68	0
s10	UPPER	FWD	no	13/72	5/20	72	0
s11	UPPER	STOP	yes	11/68	6/24	68	0

Observed scope. Both checking routes establish strictness and cooperation for all 22 original-contract graphs. Across 836 examined candidate sets, there are zero refinements and zero empty-preference fallbacks; each pair has identical raw graphs and printer text. All eleven comparisons are inactive. Thus this census supplies accepted residual controllers, including eight entries outside the source initializer, but no additional preference rescue, regression, or evidence of general effectiveness. It does not change $I_M(K)$, execute a handoff from an old controller, transfer reached private memory, or establish arbitrary-state migration. Graph construction from a prescribed state and execution of a replacement remain distinct evidence scopes.

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